Practice Plan, Lecture 1: Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 24 exercises of `Listas/pratica-lecture-01.md`

October 2026

The 24 exercises are solved in the plan's order (foundations, exam level, advanced), numbered as in the plan's tier tables. Each is cited by book, number and page; each statement is printed verbatim from the list right before its solution. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992) and Mas-Colell, Whinston & Green (1995) (MWG). Page numbers are printed pages. Notation follows the slides of Lecture 1: a game is $G = (I, \{S_i\}_{i \in I}, \{u_i\}_{i \in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of iterated elimination of strictly dominated strategies (IESDS), with every strictly dominated strategy removed in each round (slide 29), $S_i^\infty = \bigcap_k S_i^k$, and a surviving profile is an *iterated-elimination equilibrium*. In bimatrices player 1 picks rows and the first payoff is player 1's. Every matrix, dominance claim and formula is checked by the Python scripts in this folder: `check_A.py` (finite games), `check_B.py` (continuous strategies, sympy) and `check_C.py` (proofs and auctions).

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Tier 1 — Foundations

1 Tadelis Exercise 3.2

Foundations · block B1

Statement (verbatim from the list, item 1)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 3.2 "Penalty Kicks", p. 58 [PDF 77]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Penalty Kicks:** Imagine a kicker and a goalie who confront each other in a penalty kick that will determine the outcome of a soccer game. The kicker can kick the ball left or right, while the goalie can choose to jump left or right. Because of the speed of the kick, the decisions need to be made simultaneously. If the goalie jumps in the same direction as the kick, then the goalie wins and the kicker loses. If the goalie jumps in the opposite direction of the kick, then the kicker wins and the goalie loses. Model this as a normal-form game and write down the matrix that represents the game you modeled.

What is asked. Model the penalty-kick story (kicker against goalie, both choosing a side at the same time) as a normal-form game and write its payoff matrix.

Method. A normal-form game needs three objects: players, strategy sets and payoff functions (Tadelis Def. 3.3, p. 47; Lecture 1, slides 13–14). Moves are simultaneous, so a strategy is just an action (slide 17). The story only says who wins; payoffs must represent the players' preferences over outcomes (slide 18), so any numbers with "win $>$ lose" do.

Solution.

- Players: $I = \{K, G\}$ (kicker, goalie).
- Strategies: $S_K = S_G = \{L, R\}$, the side of the goal the ball is kicked to or the goalie jumps to. "Same direction" means the same side of the goal, so both players' L refer to the same post.
- Payoffs: winning is worth 1 and losing -1:

$$u_K(s_K, s_G) = \begin{cases} -1 & s_K = s_G, \\ 1 & s_K \neq s_G, \end{cases} \qquad u_G(s_K, s_G) = -u_K(s_K, s_G).$$

Kicker \ Goalie	L	R
L	$-1,\ 1$	$1,\ -1$
R	$1,\ -1$	$-1,\ 1$

Payoffs sum to zero in every cell: the game is *zero-sum* (one player's gain is the other's loss). With win $= 1$, lose $= 0$ it is constant-sum, which is strategically the same game.

Dominance (preview of the lecture's tools). For the kicker, L is better against R and worse against L; the same holds for R, and symmetrically for the goalie. No strategy is strictly or weakly

dominated, so IESDS removes nothing: $S_i^\infty = \{L, R\}$ for both. Nor is there a pure profile from which neither player wants to deviate: the loser of each cell always wants to switch. This is the structure of Matching Pennies (Tadelis 3.5); its solution needs mixed strategies (Lecture 2).

Interpretation. The point of the exercise is the modelling step: identify who moves, what each can choose without seeing the other's choice, and rank outcomes. Here pure conflict makes predictability fatal, which is why real kickers and goalies randomize.

Verification. `check_A.py` builds the matrix from the rule "goalie wins iff same side", confirms it is zero-sum, that IESDS deletes nothing and that no pure profile is stable.

Answer. $G = (\{K, G\}, \{S_K = S_G = \{L, R\}\}, \{u_K, u_G\})$ with $u_K = -1$, $u_G = 1$ if the sides match and $u_K = 1$, $u_G = -1$ otherwise (matrix above). Zero-sum; no dominated strategy, so IESDS has no bite.

2 Tadelis Exercise 3.4

Foundations · block B1

Statement (verbatim from the list, item 2)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 3.4 "Hunting", p. 58 [PDF 77]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Hunting:** Two hunters, players 1 and 2, can each choose to hunt a stag, which provides a rather large and tasty meal, or hunt a hare—also tasty, but much less filling. Hunting stags is challenging and requires mutual cooperation. If either hunts a stag alone, then the stag will get away, while hunting the stag together guarantees that the stag will be caught. Hunting hares is an individualistic enterprise that is not done in pairs, and whoever chooses to hunt a hare will catch one. The payoff from hunting a hare is 1, while the payoff to each from hunting a stag together is 3. The payoff from an unsuccessful stag hunt is 0. Represent this game as a matrix.

What is asked. Represent the two-hunter stag/hare story, with the payoffs the book gives, as a matrix. The plan also asks which payoff orderings make a game a stag hunt.

Method. Normal form (Tadelis Def. 3.3, p. 47; slides 13–14). Then compare each player's payoffs across his rows/columns, as in the dominance analysis of slide 23.

Solution. $I = \{1, 2\}$, $S_1 = S_2 = \{S, H\}$ (stag, hare). A hare hunter always catches a hare (1), whatever the other does; a stag is caught only if both hunt it (3 each); a lone stag hunter gets 0.

$1 \setminus 2$	S	H
S	3, 3	0, 1
H	1, 0	1, 1

What makes it a stag hunt. Write a player's payoffs as $a = u(S, S)$, $b = u(H, S)$, $c = u(S, H)$, $d = u(H, H)$ (own action first). The stag hunt is the symmetric game with

$$a > b \geq d > c.$$

Here $a = 3 > b = 1 = d = 1 > c = 0$. The first inequality says that hunting the stag is the best reply to a partner who hunts the stag; $d > c$ says that the hare is the best reply to a partner who hunts the hare; $b \geq d$ (hare pays at least as much whatever the partner does) makes H the safe action. Compare: if $b > a$ and $d > c$, then H would be strictly dominant and the game a prisoner's dilemma.

Dominance. For each player, S is strictly better against S ($3 > 1$) and H is strictly better against H ($1 > 0$). Neither strategy is dominated, so IESDS removes nothing. Both (S, S) and (H, H) are profiles where each player's action is a best reply to the other's (the pure Nash equilibria of Lecture 2). (S, S) Pareto-dominates every other cell (slide 26).

Interpretation. Unlike the prisoner's dilemma, nothing in individual incentives pulls players away from the efficient outcome; the problem is *confidence*: stag pays only if the partner is trusted to hunt stag too. Dominance arguments are silent here; the game is a coordination problem, the central case for Lecture 2.

Verification. `check_A.py` checks the ordering $a > b = d > c$, that IESDS deletes nothing, and that (S, S) and (H, H) are the only profiles with mutual best replies.

Answer. Matrix above: $(S, S) = (3, 3)$, $(S, H) = (0, 1)$, $(H, S) = (1, 0)$, $(H, H) = (1, 1)$. It is a stag hunt because $u(S, S) > u(H, S) \geq u(H, H) > u(S, H)$: no dominated strategies, two self-enforcing outcomes, and (S, S) is the Pareto-best one.

3 Muñoz-Garcia Ch. 1, Exercise 2

Foundations · block B1

Statement (verbatim from the list, item 3)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 1, Exercise 2 "From Extensive Form to Normal Form Representation-II", p. 4 [PDF 17]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider the extensive form game in Fig. 1.3.

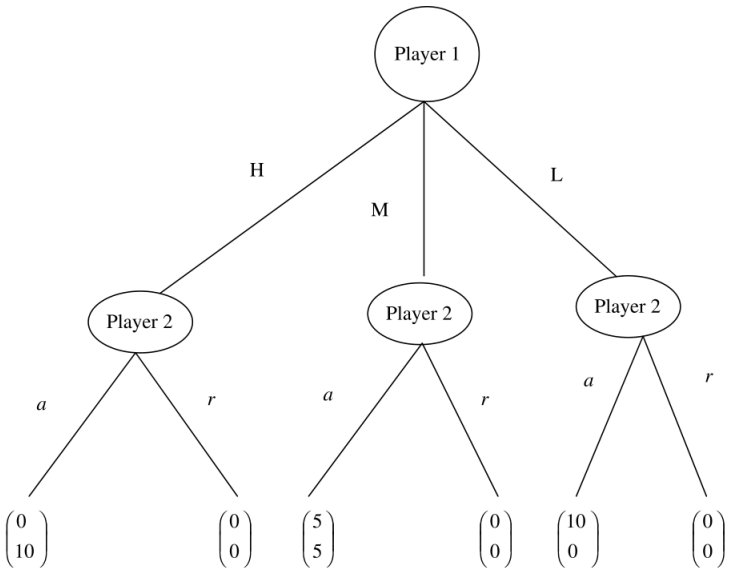


Fig. 1.3 Extensive-form game

Part (a) Describe player 1's strategy space.

Part (b) Describe player 2's strategy space.

Part (c) Take your results from parts (a) and (b) and construct a matrix representing the normal form game of this game tree.

What is asked. From the game tree of Fig. 1.3 (player 1 chooses H, M or L; player 2 observes the choice and accepts or rejects), give each player's strategy space and the normal-form matrix.

Method. A strategy is a complete contingent plan: it says what to do at every node where the player may be called to move (Lecture 1, slide 17; Tadelis §3.1, p. 47). Any game can then be written in normal form (slide 15).

Solution. **(a)** Player 1 moves once: $S_1 = \{H, M, L\}$.

(b) Player 2 moves at three nodes and can tell them apart, so a strategy is a triple xyz = (reply after H, after M, after L), each a or r:

$$S_2 = \{a, r\}^3 = \{aaa, aar, ara, arr, raa, rar, rra, rrr\}, \qquad |S_2| = 2^3 = 8.$$

Had player 2 *not* observed player 1's move (one information set), she would have only $\{a, r\}$.

(c) In cell (m, xyz) only the reply at node m matters. Leaves: $(H, a) = (0, 10)$, $(M, a) = (5, 5)$, $(L, a) = (10, 0)$, and $(0, 0)$ after any r.

	aaa	*aar*	*ara*	*arr*	*raa*	*rar*	*rra*	*rrr*
H	$0, 10$	$0, 10$	$0, 10$	$0, 10$	$0, 0$	$0, 0$	$0, 0$	$0, 0$
M	$5, 5$	$5, 5$	$0, 0$	$0, 0$	$5, 5$	$5, 5$	$0, 0$	$0, 0$
L	$10, 0$	$0, 0$	$10, 0$	$0, 0$	$10, 0$	$0, 0$	$10, 0$	$0, 0$

Comparison with the book. Our matrix agrees with the book's Fig. 1.4 (p. 5) in 23 of 24 cells. The exception is (L, aar): the book prints $(10, 0)$, but *aar* rejects after L, so the outcome is

the $(0, 0)$ leaf. This is a typo in the book; the rest of its answer (strategy sets, the remark on information sets) agrees with ours.

What dominance says (beyond the question). No strategy is strictly dominated. Player 1's H earns 0 in every column but is not *strictly* dominated, because against rrr every row earns 0. Player 2's payoff in row L is 0 whatever she does, so no plan strictly beats another in all three rows. Accepting at H and M (aaa or aar) only *weakly* dominates any plan that rejects at H or M. IESDS therefore has no bite on this matrix; the tree's sequential structure (Lecture 3) will do the work.

Interpretation. The normal form multiplies strategies: three decision nodes with two actions each give 2^3 plans, most of them differing only at nodes that are never reached. That is why the normal form of a dynamic game is large and why solving it needs more than dominance.

Verification. `check_A.py` rebuilds every cell from the six leaves, finds that only (L, aar) differs from the book's table, and checks the dominance statements above.

Answer. $S_1 = \{H, M, L\}$; $S_2 = \{a, r\}^3$ (8 plans xyz, one reply per node). Matrix above; cell (m, xyz) is the leaf of m followed by the reply to m. The book's cell (L, aar) should read $(0, 0)$, not $(10, 0)$.

4 Muñoz-Garcia Ch. 1, Exercise 5

Foundations · block B1

Statement (verbatim from the list, item 4)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 1, Exercise 5 "Prisoners' Dilemma Game", p. 8 [PDF 21]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Two individuals have been detained for a minor offense and confined in separate cells. The investigators suspect that these individuals are involved in a major crime, and separately offer each prisoner the following deal, as depicted in Fig. 1.8: if you confess while your partner doesn't, you will leave today without serving any time in jail; if you confess and your partner also confesses, you will have to serve 5 years in jail (since prosecutors probably can accumulate more evidence against each prisoner when they both confess); if you don't confess and your partner does, you will have to serve 15 years in jail (since you did not cooperate with the prosecution but your partner's confession provides the police with enough evidence against you); finally, if neither of you confess, you will have to serve one year in jail.

Prisoner 1 Prisoner 2	*Confess*	*Not confess*
Confess	-5, -5	0, -15
Not confess	-15, 0	-1, -1

Fig. 1.8 Normal-form of Prisoners' dilemma game

Part (a) Draw the prisoners' dilemma game in its extensive form representation.

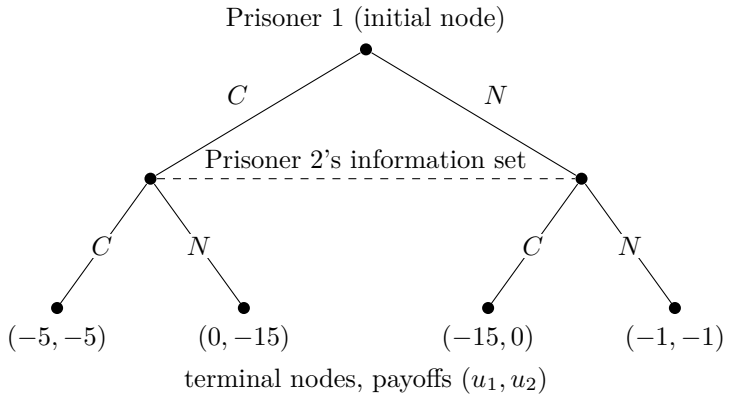
Part (b) Mark its initial node, its terminal nodes, and its information set. Why do we represent this information set in the prisoners' dilemma game in its extensive form?

Part (c) How many strategies player 1 has? What about player 2?

What is asked. (a) Draw the prisoners' dilemma of Fig. 1.8 (pp. 8–9) as a game tree; (b) mark the initial node, the terminal nodes and the information set, and explain why the information set is there; (c) count each player's strategies.

Method. Extensive form with an information set for simultaneous moves (Lecture 1, slides 15 and 17; Tadelis §3.1); dominance and Pareto optimality (slides 23–26).

Solution. Payoffs are minus the years in jail: C = confess, N = not confess.



(b) The initial node (root) is player 1's decision. The four terminal nodes carry the payoffs. The dashed line joins player 2's two decision nodes into one *information set*: she does not know which node she is at, because she decides without observing player 1's choice. This is how a tree represents simultaneous moves; drawing player 2 first instead gives the same game.

(c) A strategy assigns an action to each information set. Player 1 has one, player 2 has one (not two), so $S_1 = S_2 = \{C, N\}$: two strategies each. Without the dashed line player 2 would have $2^2 = 4$ strategies, as in item 3.

Dominance and Pareto (the lecture's lessons). For each prisoner, C strictly dominates N: $-5 > -15$ against C and $0 > -1$ against N. So (C, C) is the strict dominant-strategy equilibrium, and by slide 29 the unique IESDS survivor. But (N, N) gives $(-1, -1)$, better for both than $(-5, -5)$: the equilibrium is not Pareto optimal (slides 25–26, Lessons 1 and 2).

Interpretation. The information set encodes what a player knows when moving, and therefore what her strategies can depend on. In the dilemma, individually rational choices waste five years each.

Verification. `check_A.py` checks that C strictly dominates N for both players, that IESDS leaves only (C, C), and that (N, N) Pareto-dominates it. The book's answer (p. 10) agrees on (a)–(c); the dominance paragraph goes beyond it.

Answer. Tree above: root = prisoner 1's choice, four terminal nodes with payoffs $(-5, -5)$, $(0, -15)$, $(-15, 0)$, $(-1, -1)$, and one information set joining prisoner 2's nodes because she cannot observe prisoner 1. Each player has two strategies, $\{C, N\}$. Confess is strictly dominant; (C, C) is Pareto-dominated by (N, N).

5 Muñoz-Garcia Ch. 1, Exercise 13

Foundations · block B2

Statement (verbatim from the list, item 5)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 1, Exercise 13 "Iterated Deletion of Strictly Dominated Strategies (IDSDS)", p. 26 [PDF 39]. Reproduced verbatim (statement only; the book's worked answer follows it).

Consider the following normal form game (Fig. 1.33)

Player 1 Player 2	*Left*	*Right*
Up	-10, 10	0, 12
Middle	-12, 0	2, 2
Down	-9, 0	1, 0

Fig. 1.33 Normal-form game

(a) Find strictly dominant strategies (if any) for player 1 and for player 2.

(b) Find strictly dominated strategies (if any) for player 1 and for player 2.

(c) If you apply iterative deletion of strictly dominated strategies (IDSDS), what is the surviving strategy pair (or pairs)? Explain the steps you use in IDSDS, and why you use them.

What is asked. For the 3×2 game of Fig. 1.33: (a) strictly dominant strategies; (b) strictly dominated strategies; (c) the profiles that survive IESDS, with the steps and their justification.

Method. Definitions of strict/weak dominance and dominant strategies (Lecture 1, slide 23; Tadelis §4.1, pp. 59–63); IESDS as in slide 29 (Tadelis §4.2, p. 65).

Solution. Player 1 chooses U, M, D; player 2 chooses L, R (matrix read from the book, p. 26).

$1 \setminus 2$	L	R
U	$-10,\ 10$	$0,\ 12$
M	$-12,\ 0$	$2,\ 2$
D	$-9,\ 0$	$1,\ 0$

(a) Dominant strategies. Player 1: against L the best row is D (-9), against R it is M (2). A strictly dominant strategy would have to be the unique best row in every column, so none exists. Player 2: R is strictly better than L in rows U ($12 > 10$) and M ($2 > 0$) but equal in row D ($0 = 0$). So R *weakly* dominates L and is weakly dominant, but not strictly. No player has a strictly dominant strategy.

(b) Dominated strategies. Player 1: U is strictly dominated by D: $u_1(U, L) = -10 < -9 = u_1(D, L)$ and $u_1(U, R) = 0 < 1 = u_1(D, R)$. M is not dominated (it is the best reply to R), and neither is D (best reply to L). Player 2: L is only weakly dominated (tie in row D), R is not dominated. The only strictly dominated strategy is U.

(c) IESDS.

- Round 1: $S_1^1 = \{M, D\}$ (U removed by rationality of player 1), $S_2^1 = \{L, R\}$.
- Round 2: in the reduced game, player 2's payoffs are $(0, 0)$ from L and $(2, 0)$ from R in rows (M, D): still only weak dominance. Player 1's M and D remain best replies to R and L respectively. Nothing else is removed.

$$S^\infty = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Why this is legitimate: a rational player 1 never plays U (Lesson 1). Since rationality is common knowledge, player 2 knows it and reasons on the reduced game; there, nothing more can be ruled out by strict dominance.

Interpretation. IESDS stops at four profiles because the remaining comparisons are only weak. If we also deleted weakly dominated strategies, player 2 would drop L and player 1 would then pick M (2 against 1), giving (M, R). That stronger prediction depends on player 2 being sure that player 1 never plays D, where the two columns tie, which is why strict elimination does not make it (compare item 19).

Verification. `check_A.py` confirms that U is strictly dominated only by D, that R weakly but not strictly dominates L (before and after round 1), and that IESDS ends after one round with the four profiles above. The book's answer (pp. 26–28) agrees on (a), (b) and (c).

Answer. (a) None (player 2's R is only weakly dominant). (b) Only U, strictly dominated by D. (c) Delete U; nothing else is strictly dominated, so $S^\infty = \{M, D\} \times \{L, R\}$.

6 Tadelis Exercise 4.5

Foundations · block B3

Statement (verbatim from the list, item 6)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.5 "Iterated Elimination", p. 77 [PDF 96]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Iterated Elimination:** In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

Player 1 \ Player 2	L	C	R
U	6, 8	2, 6	8, 2
M	8, 2	4, 4	9, 5
D	8, 10	4, 6	6, 7

What is asked. Find the strategy profiles of the given 3×3 game that survive IESDS.

Method. IESDS round by round (Lecture 1, slides 28–29; Tadelis §4.2.1, p. 65): in each round remove every strategy that is strictly dominated given the opponent's surviving strategies.

Solution. Matrix as printed in the book (p. 77):

$1 \setminus 2$	L	C	R
U	6, 8	2, 6	8, 2
M	8, 2	4, 4	9, 5
D	8, 10	4, 6	6, 7

Round 1 (S^0 = full game).

- Player 1: U is strictly dominated by M: $6 < 8$, $2 < 4$, $8 < 9$. D is not dominated by M (ties $8 = 8$ and $4 = 4$), and M, D are not dominated by U.
- Player 2: her payoffs are $L = (8, 2, 10)$, $C = (6, 4, 6)$, $R = (2, 5, 7)$ in rows (U, M, D). C is beaten by L in rows U, D but not in M; it beats R in row U. No column is strictly dominated.

$S_1^1 = \{M, D\}$, $S_2^1 = \{L, C, R\}$.

Round 2. Player 2 now compares columns on rows (M, D) only: $L = (2, 10)$, $C = (4, 6)$, $R = (5, 7)$. C is strictly dominated by R ($4 < 5$, $6 < 7$); L and R are not ranked. Player 1 on all three columns: $M = (8, 4, 9)$, $D = (8, 4, 6)$: M is better only against R, so D is weakly but not strictly dominated. $S_1^2 = \{M, D\}$, $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $M = (8, 9)$, $D = (8, 6)$, a tie against L, so no strict dominance. Player 2 on (M, D): $L = (2, 10)$, $R = (5, 7)$, not ranked. The process stops.

$$S^\infty = \{M, D\} \times \{L, R\} = \{(M, L), (M, R), (D, L), (D, R)\}.$$

Interpretation. Rationality of player 1 removes U. Then player 2's knowledge that player 1 is rational removes C, which can be a best reply only to beliefs that put positive weight on U. The iterated-elimination equilibrium is a set of four profiles, not a point. Two of them, (M, R) with payoffs $(9, 5)$ and (D, L) with $(8, 10)$, are profiles where each strategy is a best reply to the other (the pure Nash equilibria of Lecture 2). Deleting the weakly dominated D as well would single out (M, R), which is a stronger and less safe assumption (see item 19).

Verification. `check_A.py` runs IESDS with the lecture's definition and finds exactly the rounds above (round 1: U; round 2: C; then stop), that U is dominated only by M and C only by R, and that D is weakly dominated by M in the final set.

Answer. Round 1: delete U (dominated by M). Round 2: delete C (dominated by R once U is gone). Nothing else is strictly dominated: $S^\infty = \{M, D\} \times \{L, R\}$, four profiles.

7 Gibbons Problem 1.2

Foundations · block B3

Statement (verbatim from the list, item 7)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §1.1 Problems, Problem 1.2, p. 48 [PDF 56]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 1.2. In the following normal-form game, what strategies survive iterated elimination of strictly dominated strategies? What are the pure-strategy Nash equilibria?

	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

What is asked. Which strategies of the given 3×3 game survive IESDS, and what are the pure-strategy Nash equilibria?

Method. IESDS (Gibbons §1.1.B, pp. 4–7; Lecture 1, slide 29). For the second question, a pure Nash equilibrium is a profile in which each player's strategy is a best reply to the other's (Gibbons §1.1.C; Lecture 2). Every Nash equilibrium survives IESDS (Gibbons Prop. B, p. 12), so we only need to search the surviving cells.

Solution. The matrix as printed (the top-left cell reads "2.0" in the scan; it is $(2, 0)$):

$1 \setminus 2$	L	C	R
T	2, 0	1, 1	4, 2
M	3, 4	1, 2	2, 3
B	1, 3	0, 2	3, 0

Round 1. Player 1: B is strictly dominated by T ($1 < 2$, $0 < 1$, $3 < 4$); it is not dominated by M ($3 > 2$ against R). T and M are best replies to R and L, so they stay. Player 2, columns over rows (T, M, B): $L = (0, 4, 3)$, $C = (1, 2, 2)$, $R = (2, 3, 0)$. C is *not* dominated yet: it beats R in row B ($2 > 0$) and L in row T. $S_1^1 = \{T, M\}$, $S_2^1 = \{L, C, R\}$.

Round 2. On rows (T, M): $C = (1, 2)$ and $R = (2, 3)$, so C is strictly dominated by R. $L = (0, 4)$ and $R = (2, 3)$ are not ranked. $S_2^2 = \{L, R\}$.

Round 3. Player 1 on (L, R): $T = (2, 4)$, $M = (3, 2)$, not ranked. Player 2 on (T, M): $L = (0, 4)$, $R = (2, 3)$, not ranked. Stop.

$$S^\infty = \{T, M\} \times \{L, R\}.$$

Pure Nash equilibria. Check the four surviving cells:

- (T, L): player 1 would switch to M ($3 > 2$). Not NE.
- (T, R): player 1 gets $4 \geq 3, 2$ in column R; player 2 gets $2 \geq 0, 1$ in row T. NE.
- (M, L): player 1 gets $3 \geq 2, 1$; player 2 gets $4 \geq 2, 3$. NE.
- (M, R): player 1 would switch to T ($4 > 2$). Not NE.

The pure Nash equilibria are (T, R) and (M, L).

Interpretation. Note the order: C is a sensible reply only to beliefs that put weight on B; once player 2 knows player 1 is rational, C goes. IESDS leaves four profiles; Nash equilibrium, which requires beliefs to be correct, cuts them to two. This previews Lecture 2.

Verification. `check_A.py` reproduces the rounds (round 1: B; round 2: C), checks that C is not dominated in the full game, and finds by brute force over all nine cells that the pure Nash equilibria are exactly (T, R) and (M, L). Kumar's unofficial Gibbons manual (Answer 1.2, not in `Livros/`) gives the same deletions and equilibria; it writes the survivors as "(T,M) and (L,R)", meaning $\{T, M\}$ for player 1 and $\{L, R\}$ for player 2.

Answer. IESDS: delete B (by T), then C (by R); survivors $\{T, M\} \times \{L, R\}$. Pure Nash equilibria: (T, R) with payoffs $(4, 2)$ and (M, L) with payoffs $(3, 4)$.

8 Muñoz-Garcia Ch. 1, Exercise 9

Foundations · block B3

Statement (verbatim from the list, item 8)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 1, Exercise 9 "Applying IDSDS in the Battle of the Sexes Game", p. 17 [PDF 30]. Reproduced verbatim (statement only; the book's worked answer follows it).

> A husband and a wife are leaving work, and do not remember which event they are attending to tonight. Both of them, however, remember that last night's argument was about attending either the football game (the most preferred event for the husband) or the opera (the most preferred event for the wife). To make matters worse, their cell phones broke, so they cannot call each other to confirm which event they are attending to. As a consequence, they must simultaneously and independently decide whether to attend to the football game or the opera.

Husband *Wife*	*Football*	*Opera*
Football	3, 1	0, 0
Opera	0, 0	1, 3

Fig. 1.22 Normal-form representation of the Battle of the Sexes game

> The payoff matrix in Fig. 1.22, describes the preference of the husband (wife) for the football game (opera, respectively). Payoffs also indicate that both players prefer to be together rather than being alone (even if they are alone at their most preferred event). Apply IDSDS. What is the most precise prediction about how this game will be played when you apply IDSDS?

What is asked. Apply IESDS to the husband–wife Battle of the Sexes of Fig. 1.22 (p. 18) and state the most precise prediction it gives.

Method. IESDS (Lecture 1, slide 29); evaluation criteria for solution concepts, in particular uniqueness (slide 22; Tadelis §4.2.3, pp. 67–68).

Solution.

Husband \ Wife	F	O
F	3, 1	0, 0
O	0, 0	1, 3

- Husband: $u_H(F,F) = 3 > 0 = u_H(O,F)$ but $u_H(O,O) = 1 > 0 = u_H(F,O)$. Each strategy is strictly better against one of the wife's strategies, so neither is dominated, strictly or weakly.
- Wife: $u_W(F,F) = 1 > 0 = u_W(F,O)$ but $u_W(O,O) = 3 > 0 = u_W(O,F)$. The same holds.

Round 1 removes nothing, so the process stops at once: $S_i^1 = S_i^0$ and

$$S^\infty = \{F,O\} \times \{F,O\} = \{(F,F),(F,O),(O,F),(O,O)\}.$$

Interpretation. IESDS has no bite here: every profile is an iterated-elimination equilibrium, including the miscoordinated (F,O) and (O,F), which no one wants. The reason is structural. In a coordination game each strategy is the best reply to *something*, so rationality and common knowledge of rationality cannot rule anything out. IESDS exists for every game, but it often fails on uniqueness (slide 22). What is missing is a theory of what each player *expects* the other to do. Nash equilibrium supplies it (Lecture 2) and selects (F,F) and (O,O) in pure strategies. The same failure occurs in the stag hunt (item 2) and in matching pennies (item 1).

Verification. `check_A.py` confirms that no strategy is strictly or weakly dominated and that IESDS returns the full game. The book's answer (p. 18) agrees: no strictly (or weakly) dominated strategies and all four profiles survive.

Answer. No strategy is dominated for either player (each is the strict best reply to one of the other's choices), so IESDS deletes nothing and all four profiles survive. The "most precise prediction" is that anything can happen; IESDS has no bite in coordination games.

Tier 2 — Exam level

9 Tadelis Exercise 3.6

Exam · block B1

Statement (verbatim from the list, item 9)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 3.6 "Price Competition", p. 58 [PDF 77]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Price Competition:** Imagine a market with demand $p(q) = 100 - q$. There are two firms, 1 and 2, and each firm i has to simultaneously choose its price p_i. If $p_i < p_j$, then firm i gets all of the market while no one demands the good of firm j. If the prices are the same then both firms split the market demand equally. Imagine that there are no costs to produce any quantity of the good. (These are two large dairy farms, and the product is manure.) Write down the normal form of this game.

What is asked. Write the normal form of a price game between two zero-cost firms facing demand $p = 100 - q$, where the cheaper firm takes the whole market and equal prices split it.

Method. Normal form with continuous strategy sets and piecewise payoffs, built like Tadelis's Cournot example (§3.1.2, p. 49) and the lecture's definition (slides 13–14, 18).

Solution. Inverting the demand curve, a firm that serves the market at price p sells $D(p) = \max\{0,\, 100 - p\}$ (no one buys at $p \geq 100$). Costs are zero, so revenue is profit.

- Players: $I = \{1, 2\}$.
- Strategy sets: $S_i = [0, \infty)$, prices (one may cap them at 100 without loss, since higher prices sell nothing).
- Payoffs: for $i \neq j$,

$$u_i(p_i, p_j) = \begin{cases} p_i \max\{0, 100 - p_i\} & \text{if } p_i < p_j, \\ \frac{1}{2}\, p_i \max\{0, 100 - p_i\} & \text{if } p_i = p_j, \\ 0 & \text{if } p_i > p_j. \end{cases}$$

This triple $G = (\{1, 2\}, \{[0, \infty)\}_i, \{u_i\}_i)$ is the normal form. It cannot be drawn as a finite matrix: each player has a continuum of strategies, and the payoff jumps at $p_i = p_j$.

Dominance (beyond the question). Against $p_j = 0$ every price earns 0 (a higher price sells nothing; a tie at 0 earns nothing). So no price is *strictly* better than another against every p_j: no strategy is strictly dominated and IESDS has no bite. Weak dominance does bite. The monopoly price maximizes $p(100 - p)$ at $p^m = 50$ ($100 - 2p = 0$), and p^m earns at least as much as any $p_i \geq 100$ (which always earns 0) or $p_i = 0$, and strictly more against $p_j > 50$. Lecture 2's Bertrand equilibrium, $p_1 = p_2 = 0$, will therefore use weakly dominated strategies.

Interpretation. Even a simple pricing story gives a discontinuous payoff function: undercutting by a cent doubles the quantity sold. That discontinuity drives the Bertrand result (Lecture 2). It is also why the normal form must specify the tie-breaking rule.

Verification. `check_B.py` builds the payoff on an integer price grid $0, \ldots, 120$, confirms that every price earns 0 against $p_j = 0$, that grid IESDS deletes nothing, that $p = 50$ weakly dominates $p = 0$ and every $p \geq 100$, and that 50 solves the first-order condition (sympy).

Answer. $I = \{1, 2\}$, $S_i = [0, \infty)$, and $u_i(p_i, p_j) = p_i \max\{0, 100 - p_i\}$ if $p_i < p_j$, half of that if $p_i = p_j$, and 0 if $p_i > p_j$. No strategy is strictly dominated; prices 0 and ≥ 100 are weakly dominated by 50.

10 Tadelis Exercise 3.7

Exam · block B1

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 3.7 "Public Good Contribution", p. 58 [PDF 77]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Public Good Contribution:** Three players live in a town, and each can choose to contribute to fund a streetlamp. The value of having the streetlamp is 3 for each player, and the value of not having it is 0. The mayor asks each player to contribute either 1 or nothing. If at least two players contribute then the lamp will be erected. If one player or no players contribute

then the lamp will not be erected, in which case any person who contributed will not get his money back. Write down the normal form of this game.

What is asked. Write the normal form of the three-player streetlamp game (contribute 1 or nothing; the lamp, worth 3 to everyone, is built iff at least two contribute; contributions are not refunded).

Method. Normal form with $n = 3$ players (slides 13–14); a three-player game is drawn as one matrix for players 1 and 2 per strategy of player 3 (Tadelis §3.1).

Solution. Let $s_i = 1$ if player i contributes and $s_i = 0$ otherwise.

- Players: $I = \{1, 2, 3\}$; strategies $S_i = \{0, 1\}$, so $|S| = 2^3 = 8$ profiles.
- Payoffs: the lamp is built iff $s_1 + s_2 + s_3 \geq 2$, so

$$u_i(s_1, s_2, s_3) = 3 \cdot \mathbf{1}\{s_1 + s_2 + s_3 \geq 2\} - s_i.$$

Payoffs (u_1, u_2, u_3); player 1 picks the row, player 2 the column, player 3 the matrix:

Player 3: $s_3 = 0$

$1 \setminus 2$	0	1
0	0, 0, 0	0, −1, 0
1	−1, 0, 0	2, 2, 3

Player 3: $s_3 = 1$

$1 \setminus 2$	0	1
0	0, 0, −1	3, 2, 2
1	2, 3, 2	2, 2, 2

Dominance (beyond the question). Player i's best action depends on how many *others* contribute, $m \in \{0, 1, 2\}$:

$$u_i(1; m) - u_i(0; m) = \begin{cases} -1 & m = 0 \text{ (lamp fails either way)}, \\ 3 - 1 = 2 & m = 1 \text{ (player } i \text{ is pivotal)}, \\ -1 & m = 2 \text{ (lamp built anyway)}. \end{cases}$$

Neither action is dominated, so IESDS removes nothing. The profiles where every player is best-replying are "nobody contributes" and the three profiles with exactly two contributors.

Interpretation. Contributing pays only when one is pivotal. That is the free-rider problem of public goods: with two others contributing, a player keeps her coin. The game has both coordination (no lamp vs. lamp) and conflict (who pays).

Verification. `check_A.py` evaluates the payoff formula on all eight profiles (it matches the matrices), confirms that IESDS deletes nothing for any player, and finds the four profiles with mutual best replies by brute force.

Answer. $I = \{1, 2, 3\}$, $S_i = \{0, 1\}$, $u_i(s) = 3 \cdot \mathbf{1}\{\sum_j s_j \geq 2\} - s_i$; matrices above, one per choice of player 3. No strategy is dominated: contributing pays iff exactly one other player contributes.

11 Tadelis Exercise 4.1

Exam · block B2

Statement (verbatim from the list, item 11)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.1, p. 76 [PDF 95]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Prove Proposition 4.1:** If the game $\Gamma = \langle N, \{S_i\}_{i=1}^n, \{v_i\}_{i=1}^n \rangle$ has a strictly dominant strategy equilibrium s^D, then s^D is the unique dominant strategy equilibrium.

What is asked. Prove Tadelis's Proposition 4.1 (p. 61): a strict dominant-strategy equilibrium, if it exists, is the unique dominant-strategy equilibrium.

Method. Direct proof by contradiction from the definitions (Lecture 1, slides 23 and 25; Tadelis Defs. 4.1–4.3, pp. 60–61). "Dominant-strategy equilibrium" is read in the widest sense: a profile in which every player uses a dominant strategy, strict *or* weak. Tadelis defines weak dominance with "$\geq$ for all s_{-i}" (Remark, p. 63); the lecture adds "$>$ for some s_{-i}". The proof works under both readings because it only uses the "$\geq$" part.

Solution. **Claim.** Let $s^D \in S$ be a strict dominant-strategy equilibrium of $G = (I, \{S_i\}, \{u_i\})$: for every $i \in I$,

$$u_i(s_i^D, s_{-i}) > u_i(s_i', s_{-i}) \qquad \forall s_i' \in S_i \setminus \{s_i^D\},\ \forall s_{-i} \in S_{-i}. \tag{1}$$

If $s^W \in S$ is a profile in which every player uses a (strictly or weakly) dominant strategy, then $s^W = s^D$.

Proof. Suppose not. Then there is a player i with $s_i^W \neq s_i^D$. Because s_i^W is (at least weakly) dominant, it does at least as well as every strategy, in particular s_i^D:

$$u_i(s_i^W, s_{-i}) \geq u_i(s_i^D, s_{-i}) \qquad \forall s_{-i} \in S_{-i}. \tag{2}$$

Take any $s_{-i} \in S_{-i}$ (the set is nonempty). Applying (1) with $s_i' = s_i^W$ gives $u_i(s_i^D, s_{-i}) > u_i(s_i^W, s_{-i})$, which contradicts (2) at that s_{-i}. Hence $s_i^W = s_i^D$ for every i, that is, $s^W = s^D$. □

What the proof uses. Only the definitions; no finiteness. Two by-products:

(i) A player has at most one strictly dominant strategy (take s_i^W strictly dominant in the argument).

(ii) If the dominance were only weak in (1), the argument fails: two strategies can each be "$\geq$" the other everywhere when they tie everywhere. That is exactly Exercise 4.2(b) (item 12).

The lecture's other proposition (slide 29; Tadelis Prop. 4.2, p. 68) adds that s^D is also the unique IESDS survivor: in round 1 every $s_i' \neq s_i^D$ is strictly dominated by s_i^D.

Interpretation. When it exists, strict dominance gives a unique prediction and asks very little of the players (only their own rationality). The cost is existence: most games have no strictly dominant strategies (item 8).

Verification. `check_C.py` draws 20 000 random games (2×2, 2×3, 3×2, payoffs in $\{0,\dots,4\}$). Whenever both players have a strictly dominant strategy, it checks that the strictly dominant strategy is the only "$\geq$-dominant" one for each player and the only IESDS survivor.

Answer. If $s_i^W \neq s_i^D$ for some i, weak dominance of s_i^W gives $u_i(s_i^W, s_{-i}) \geq u_i(s_i^D, s_{-i})$ for all s_{-i}, while strict dominance of s_i^D gives the reverse strict inequality, a contradiction. So any dominant-strategy equilibrium equals s^D.

12 Tadelis Exercise 4.2

Exam · block B2

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.2 "Weak Dominance", p. 76 [PDF 95]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Weak Dominance:** We call the strategy profile $s^W \in S$ a **weakly dominant strategy equilibrium** if $s_i^W \in S_i$ is a weakly dominant strategy for all $i \in N$, that is, if $v_i(s_i, s_{-i}) \geq v_i(s_i', s_{-i})$ for all $s_i' \in S_i$ and for all $s_{-i} \in S_{-i}$.
>
> a. Provide an example of a game in which there is no weakly dominant strategy equilibrium.
>
> b. Provide an example of a game in which there is more than one weakly dominant strategy equilibrium.

What is asked. Using Tadelis's definition of a weakly dominant strategy equilibrium, give (a) a game with no such equilibrium and (b) a game with more than one.

Method. Definitions. Tadelis calls s_i^W weakly dominant if $u_i(s_i^W, s_{-i}) \geq u_i(s_i', s_{-i})$ for all s_i' and all s_{-i}, with no strictness required (statement of 4.2 and Remark, p. 63). The lecture's weak dominance (slide 23) also asks for a strict inequality at some s_{-i}. Which definition is used matters for (b).

Solution. **(a) No weakly dominant strategy equilibrium.** Matching pennies (Tadelis 3.5), or the penalty kicks of item 1:

$1 \setminus 2$	H	T
H	$1,\ -1$	$-1,\ 1$
T	$-1,\ 1$	$1,\ -1$

For player 1, H does worse than T against T ($-1 < 1$), and T does worse than H against H. So neither satisfies "$\geq$ every other strategy for every s_2". The same holds for player 2. No player has a weakly dominant strategy, so there is no weakly dominant strategy equilibrium. (The Battle of the Sexes, item 8, works too.)

(b) More than one. Make one player indifferent between two strategies and give the other a dominant strategy:

$1 \setminus 2$	L	R
U	2, 3	0, 1
D	2, 1	0, 0

Player 1's payoffs from U and D coincide column by column ($2 = 2$, $0 = 0$), so each satisfies Tadelis's "$\geq$" condition against the other: both are weakly dominant. For player 2, L strictly dominates R ($3 > 1$, $1 > 0$). Hence (U, L) and (D, L) are both weakly dominant strategy equilibria. They are not equivalent: player 2 gets 3 in one and 1 in the other. This is the same idea as the MWG solutions manual's example for MWG Exercise 8.B.2(b).

The role of the definition. Under the lecture's definition, (b) is impossible. If s_i and s_i' were both weakly dominant, s_i would weakly dominate s_i', so $u_i(s_i, s_{-i}) > u_i(s_i', s_{-i})$ for some s_{-i}, while s_i' weakly dominating s_i requires $u_i(s_i', s_{-i}) \geq u_i(s_i, s_{-i})$ at that same s_{-i}. Contradiction. So each player has at most one weakly dominant strategy and there is at most one weakly dominant-strategy equilibrium. In general, as MWG Exercise 8.B.2(a) shows, two strategies that are both weakly dominant in the "$\geq$" sense give equal payoffs against every s_{-i}. Multiplicity comes only from such ties.

Interpretation. Weak dominance is a weaker requirement, so an equilibrium in weakly dominant strategies is easier to have (the second-price auction, item 23) but need not be unique. The non-uniqueness here is harmless for the indifferent player but not for the other one.

Verification. `check_C.py` checks that no strategy in matching pennies is "$\geq$-dominant", that U and D are both "$\geq$-dominant" and L strictly dominant in the example, that player 2 prefers (U, L), and, on 20 000 random games, that under the lecture's definition no player ever has two weakly dominant strategies.

Answer. (a) Matching pennies: every strategy is strictly worse than the other against some opponent strategy, so no weakly dominant strategy exists. (b) In the 2×2 game above, U and D tie for player 1 and L is strictly dominant for player 2, so (U, L) and (D, L) are both weakly dominant strategy equilibria. (b) needs Tadelis's "$\geq$ only" definition; under the lecture's definition it cannot happen.

13 MWG Exercise 8.B.1

Exam · block B2

Statement (verbatim from the list, item 13)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.B.1^A, p. 262 [PDF 142]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

> There are I firms in an industry. Each can try to convince Congress to give the industry a subsidy. Let h_i denote the number of hours of effort put in by firm i, and let $c_i(h_i) = w_i(h_i)^2$, where w_i is a positive constant, be the cost of this effort to firm i. When the effort levels of the firms are $(h_1, \ldots, h_I)$, the value of the subsidy that gets approved is $\alpha \sum_i h_i + \beta(\prod_i h_i)$, where α and β are constants.
>
> Consider a game in which the firms decide simultaneously and independently how many hours they will each devote to this effort. Show that each firm has a strictly dominant strategy if and only if $\beta = 0$. What is firm i's strictly dominant strategy when this is so?

What is asked. I firms choose lobbying hours $h_i \geq 0$ at cost $w_i h_i^2$; the subsidy is worth $\alpha \sum_i h_i + \beta \prod_i h_i$. Show that each firm has a strictly dominant strategy if and only if $\beta = 0$, and find it.

Method. Strict dominance (MWG Def. 8.B.1, p. 237; Lecture 1, slide 23) in a game with continuous strategies. With a strictly concave payoff, the best reply is unique. So a strategy is strictly dominant exactly when it is the best reply to *every* profile of the others.

Solution. We take firm i's payoff to be the value of the subsidy minus its own cost (as the MWG solutions manual does), with $I \geq 2$, $w_i > 0$. Write $P_{-i} = \prod_{j \neq i} h_j \geq 0$:

$$u_i(h_i, h_{-i}) = (\alpha + \beta P_{-i})\, h_i - w_i h_i^2 + \alpha \textstyle\sum_{j \neq i} h_j.$$

The last term does not depend on h_i. Also $\partial^2 u_i / \partial h_i^2 = -2w_i < 0$.

Step 1: best reply. Strict concavity on $[0, \infty)$ gives the unique maximizer

$$BR_i(h_{-i}) = \max\Big\{0,\ \frac{\alpha + \beta P_{-i}}{2w_i}\Big\}.$$

When positive, it solves the first-order condition $\alpha + \beta P_{-i} - 2w_i h_i = 0$.

Step 2: dominance $\Leftrightarrow$ constant best reply. If h_i^* is strictly dominant, it is a best reply to every h_{-i}, so BR_i is the constant h_i^*. Conversely, if $BR_i(h_{-i}) = h_i^*$ for all h_{-i}, uniqueness of the maximizer gives $u_i(h_i^*, h_{-i}) > u_i(h_i, h_{-i})$ for every $h_i \neq h_i^*$ and every h_{-i}: h_i^* is strictly dominant. With $I \geq 2$, P_{-i} takes every value in $[0, \infty)$ (set each $h_j = P^{1/(I-1)}$).

Step 3: when is BR_i constant in $P_{-i} \in [0, \infty)$?

- $\beta = 0$: $BR_i = \max\{0, \alpha/(2w_i)\}$, constant. *Dominant.* For $\alpha > 0$,

$$u_i(h_i^*, h_{-i}) - u_i(h_i, h_{-i}) = w_i\,(h_i - h_i^*)^2 > 0 \quad (h_i \neq h_i^*), \qquad h_i^* = \frac{\alpha}{2w_i}.$$

- $\beta > 0$: $(\alpha + \beta P)/(2w_i) \to \infty$ as $P \to \infty$. Not constant: no dominant strategy (more effort by others raises the return to own effort).
- $\beta < 0$, $\alpha > 0$: $BR_i = \alpha/(2w_i) > 0$ at $P = 0$ but $BR_i = 0$ for $P \geq \alpha/|\beta|$. Not constant.
- $\beta < 0$, $\alpha \leq 0$: $\alpha + \beta P \leq 0$ for all $P \geq 0$, so $BR_i \equiv 0$. *Dominant*, although $\beta \neq 0$: for $h_i > 0$, $u_i(0, h_{-i}) - u_i(h_i, h_{-i}) = -(\alpha + \beta P_{-i})h_i + w_i h_i^2 > 0$.

So "strictly dominant $\Leftrightarrow \beta = 0$" holds whenever $\alpha > 0$, the case in which lobbying is productive and the dominant strategy is interior. At the corner, when $\alpha \leq 0$ and $\beta < 0$, lobbying never pays, so $h_i = 0$ is strictly dominant for any such β.

Interpretation. $\beta \neq 0$ makes efforts interact: with $\beta > 0$ they are complements (others' hours raise my marginal return), so my best choice depends on what I expect them to do and no strategy is dominant. With $\beta = 0$ each firm's problem is separable: it equates its own marginal benefit α to its marginal cost $2w_i h_i$, whatever the others do. Even then the outcome is not efficient for the industry: each firm ignores that its hours also benefit the other firms' subsidy, the same wedge as in the prisoner's dilemma (slide 26).

Verification. `check_B.py` derives the best reply and the second derivative with sympy, checks the identity $u_i(h^*) - u_i(h) = w_i(h - h^*)^2$ for $\beta = 0$, and tests nine sign combinations of (α, β) on values of P_{-i} from 0 to 10^3: the best reply is constant exactly when $\beta = 0$ or ($\beta < 0$, $\alpha \leq 0$).

Comparison with the MWG solutions manual (p. 8-1): it agrees on the first-order condition, the "iff $\beta = 0$" and $h_i = \alpha/(2w_i)$, but it ignores the constraint $h_i \geq 0$. Its "only if" therefore misses the corner case $\alpha \leq 0$, $\beta < 0$, and for $\alpha < 0$, $\beta = 0$ its formula gives a negative number of hours (the dominant strategy is then $h_i = 0$).

Answer. Each firm's payoff is strictly concave with best reply $\max\{0, (\alpha + \beta \prod_{j \neq i} h_j)/(2w_i)\}$, and a strategy is strictly dominant iff this best reply is constant. For $\alpha > 0$ that happens iff $\beta = 0$, and the strictly dominant strategy is $h_i^* = \alpha/(2w_i)$. (Corner: with $\alpha \leq 0$ the dominant strategy is $h_i = 0$, also when $\beta < 0$.)

14 Muñoz-Garcia Ch. 1, Exercise 12

Exam · block B3

Statement (verbatim from the list, item 14)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 1, Exercise 12 "IDSDS and Rationality", p. 23 [PDF 36]. Reproduced verbatim (statement only; the book's worked answer follows it).

Consider the following 3×3 matrix (Fig. 1.30):

Player 1	Player 2	F	C	B
F		0, 5	2, 4	2, 3
C		2, 3	0, 5	3, 2
B		5, 0	2, 4	2, 3

Fig. 1.30 Normal-form representation of a three-players game

a. Does any player have a strictly dominant strategy?

b. Does any player have a strictly dominated strategy?

c. Use your results from part (b) to delete players' strictly dominated strategies assuming rationality, but not assuming common knowledge of rationality. [Hint: This means that every player can delete the strategies he finds to be strictly dominated but cannot put himself in the shoes of his opponent to delete the strategies his opponent finds to be strictly dominated.]

d. Use your results from part (b) to delete players' strictly dominated strategies assuming common knowledge of rationality. [Hint: This means that players can put themselves in the shoes of their opponents to delete strategies each other finds to be strictly dominated, as in the standard approach of IDSDS.]

What is asked. For the 3×3 game of Fig. 1.30: (a) strictly dominant strategies; (b) strictly dominated strategies; (c) what each player deletes under rationality alone; (d) what survives under common knowledge of rationality (IESDS).

Method. Dominance (slide 23) and the epistemic reading of IESDS (slide 28; Tadelis §4.2.1, p. 65). Round 1 of IESDS uses only each player's own rationality. Round 2 uses "each knows the other is rational", and so on.

Solution. Matrix from the book (p. 23):

$1 \setminus 2$	F	C	B
F	0, 5	2, 4	2, 3
C	2, 3	0, 5	3, 2
B	5, 0	2, 4	2, 3

(a) Player 1's best rows are B against F (5), F or B against C (tie at 2), and C against B (3). No row is the unique best everywhere: no strictly dominant strategy. Player 2's best columns are F in row F (5), C in row C (5), C in row B (4): none is best everywhere. Neither player has a strictly dominant strategy.

(b) Player 1: $F = (0, 2, 2)$, $C = (2, 0, 3)$, $B = (5, 2, 2)$ over columns (F, C, B). B and C are the unique best replies to columns F and B, so they are not dominated; F ties with B against columns C and B and beats C against column C, so it is not strictly dominated either (it is weakly dominated by B). No row is strictly dominated. Player 2: $F = (5, 3, 0)$, $C = (4, 5, 4)$, $B = (3, 2, 3)$ over rows (F, C, B). B is strictly dominated by C: $3 < 4$, $2 < 5$, $3 < 4$. F and C are not dominated (each is the best reply to some row). The only strictly dominated strategy is player 2's B.

(c) Rationality, no common knowledge. Each player deletes his own strictly dominated strategies but does not use the fact that the other does the same. Player 2 drops B; player 1 has nothing to drop and still evaluates his rows against all three columns:

$$\{F, C, B\} \times \{F, C\} = \{(F, F), (F, C), (C, F), (C, C), (B, F), (B, C)\}.$$

(d) Common knowledge of rationality. Player 1 knows player 2 is rational, so he compares rows on columns $\{F, C\}$ only: $F = (0, 2)$, $C = (2, 0)$, $B = (5, 2)$. Now C is strictly dominated by B ($2 < 5$, $0 < 2$); F is only weakly dominated by B (tie against C). Round 3: player 2 compares $F = (5, 0)$ and $C = (4, 4)$ on rows (F, B); neither dominates. Player 1 compares $F = (0, 2)$ and $B = (5, 2)$; only weak. Stop:

$$S^{\infty} = \{F, B\} \times \{F, C\} = \{(F, F), (F, C), (B, F), (B, C)\}.$$

Interpretation. Rationality alone removes only strategies that are bad against *everything*. Player 1's C is a good reply to B, so it can only be discarded by someone who knows that player 2 will not play B, that is, by someone who uses player 2's rationality. Each extra layer of "I know that you know" can remove more strategies; here two layers suffice.

Verification. `check_A.py` checks the dominance statements in (a)–(b), that IESDS removes B for player 2 in round 1 and C for player 1 in round 2, and that F is only weakly dominated by B in the final game. The book's answer (pp. 23–26) agrees with (a)–(d). Its text for (b) and (d) describes the comparisons loosely (for example, it says player 1 deletes C because his payoff "is higher from B ... when player 2 plays F"), but the deletions and the final sets are the same as ours.

Answer. (a) None. (b) Only player 2's B (strictly dominated by C). (c) Rationality alone: $\{F, C, B\} \times \{F, C\}$, six profiles. (d) With common knowledge, player 1 then drops C (dominated by B on $\{F, C\}$): $S^{\infty} = \{F, B\} \times \{F, C\}$, four profiles.

15 Tadelis Exercise 4.6

Exam · block B3

Statement (verbatim from the list, item 15)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.6 "Roommates", p. 77 [PDF 96]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Roommates:** Two roommates each need to choose to clean their apartment, and each can choose an amount of time $t_i \geq 0$ to clean. If their choices are t_i and t_j, then player i's payoff is given by $(10 - t_j)t_i - t_i^2$. (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)
>
> a. What is the best response correspondence of each player i?
>
> b. Which choices survive one round of IESDS?
>
> c. Which choices survive IESDS?

What is asked. Two roommates choose cleaning times $t_i \geq 0$ with payoff $u_i = (10 - t_j)\, t_i - t_i^2$. Find (a) the best-response correspondence, (b) the choices that survive one round of IESDS, and (c) those that survive IESDS.

Method. IESDS with continuous strategies by bounding best replies, exactly as in Tadelis's Cournot example (§4.2.2, pp. 65–67, Fig. 4.1). Tool: for a payoff that is a concave parabola in own action, an action outside the range of possible best replies is strictly dominated by the nearest end of that range.

Solution. **(a) Best response.** u_i is strictly concave in t_i ($\partial^2 u_i / \partial t_i^2 = -2$). The first-order condition $10 - t_j - 2t_i = 0$ and the constraint $t_i \geq 0$ give the unique best reply

$$BR_i(t_j) = \max\left\{0,\ \frac{10 - t_j}{2}\right\} = \max\left\{0,\ 5 - \frac{t_j}{2}\right\}.$$

It is a function (single-valued) and decreasing in t_j: cleaning is a strategic substitute.

Dominance lemma. Write $x = (10 - t_j)/2$, so $u_i(t, t_j) = t(2x - t)$. For any two times a, t,

$$u_i(a, t_j) - u_i(t, t_j) = (t - a)\,(t + a - 2x). \tag{1}$$

If $t > a \geq x$ the right side is positive; if $t < a \leq x$ it is positive too. So if every possible x (given the opponent's surviving set) lies in $[\underline{a}, \bar{a}]$, then every $t > \bar{a}$ is strictly dominated by $\bar{a}$ and every $t < \underline{a}$ by $\underline{a}$. Conversely, a t that is the unique best reply to some surviving t_j is not strictly dominated.

(b) One round. With $t_j \in [0, \infty)$, the best reply ranges over $[0, 5]$ ($BR = 5$ at $t_j = 0$, $BR = 0$ for $t_j \geq 10$). By (1) with $\bar{a} = 5$, every $t_i > 5$ is strictly dominated by $t_i = 5$. Every $t_i \in [0, 5]$ is the best reply to $t_j = 10 - 2t_i \in [0, 10]$, so it survives. Hence $S_i^1 = [0, 5]$.

(c) Iterating. If $S_j^k = [L_k, U_k]$, the surviving best replies form $[BR(U_k), BR(L_k)]$, so

$$L_{k+1} = 5 - \tfrac{U_k}{2}, \qquad U_{k+1} = 5 - \tfrac{L_k}{2}, \qquad (L_1, U_1) = (0, 5).$$

k	1	2	3	4	5	∞
S_i^k	$[0, 5]$	$[2.5, 5]$	$[2.5, 3.75]$	$[3.125, 3.75]$	$[3.125, 3.4375]$	$\{10/3\}$

The width halves each round: $U_{k+1} - L_{k+1} = \frac{1}{2}(U_k - L_k)$, so $U_k - L_k = 5/2^{k-1} \to 0$. The midpoint M_k satisfies $M_{k+1} = 5 - M_k/2$, whose fixed point is $10/3$, and

$$M_k - \tfrac{10}{3} = (-\tfrac{1}{2})^{k-1}(M_1 - \tfrac{10}{3}) = (-\tfrac{1}{2})^{k-1}(-\tfrac{5}{6}) \to 0.$$

Therefore $S_i^\infty = \bigcap_k S_i^k = \{10/3\}$ for both roommates: IESDS predicts the unique profile $t_1 = t_2 = 10/3$, the point where $t = 5 - t/2$.

Interpretation. Each round uses one more layer of knowledge: round 1, "I am rational" (never clean more than 5 hours); round 2, "I know you are rational" (you clean at most 5, so I clean at least 2.5); and so on. Each player gets $u_i = (10 - 10/3)(10/3) - (10/3)^2 = 100/9$. Both cleaning $t = 2.5$ would give $(7.5)(2.5) - 6.25 = 12.5 > 100/9$: the IESDS outcome over-cleans relative to the joint optimum. Each roommate ignores that his cleaning lowers the value of the other's.

Verification. `check_B.py` checks the best reply and the factorization (1) with sympy, runs the interval recursion for 40 rounds in exact arithmetic (width $5/2^{k-1}$ and the midpoint formula at every round), and, as a second route, runs brute-force IESDS on the grid $\{0, \frac{1}{6}, \ldots, 10\}$, which leaves exactly $t_1 = t_2 = 10/3$.

Answer. (a) $BR_i(t_j) = \max\{0, (10 - t_j)/2\}$. (b) $S_i^1 = [0, 5]$: any $t_i > 5$ is strictly dominated by 5. (c) Rounds give $[2.5, 5]$, $[2.5, 3.75]$, $[3.125, 3.75]$, …, with widths $5/2^{k-1}$; only $t_1 = t_2 = 10/3$ survives IESDS.

16 Tadelis Exercise 4.7

Exam · block B3

Statement (verbatim from the list, item 16)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.7 "Campaigning", pp. 77–78 [PDF 96–97]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Campaigning:** Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign $[P]$), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign $[B]$), or finally focus only on attacking one's opponent (call this a negative campaign $[N]$). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability $\pi \in [0, 1]$, then his payoff is π. The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:
>
> If both choose the same campaign then each wins with probability 0.5.
>
> If candidate i uses a positive campaign while $j \neq i$ uses a balanced one then i loses for sure.
>
> If candidate i uses a positive campaign while $j \neq i$ uses a negative one then i wins with probability 0.3.
>
> If candidate i uses a negative campaign while $j \neq i$ uses a balanced one then i wins with probability 0.6.
>
> a. Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)

b. Write down the game in matrix form.

c. What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

What is asked. Two candidates choose a positive (P), balanced (B) or negative (N) campaign; payoffs are win probabilities given by the book's rules. (a) Model it as a normal-form game; (b) write the matrix; (c) describe each stage of IESDS and say whether it gives a clear prediction.

Method. Normal form (slides 13–14); IESDS (slide 29; Tadelis §4.2).

Solution. **(a)** $I = \{1, 2\}$, $S_1 = S_2 = \{P, B, N\}$. Let $\pi(s_i, s_j)$ be the probability that the candidate playing s_i wins against s_j. The book's rules give

$$\pi(s, s) = 0.5, \quad \pi(P, B) = 0, \quad \pi(P, N) = 0.3, \quad \pi(N, B) = 0.6.$$

The other three entries follow because exactly one candidate wins: $\pi(B, P) = 1$, $\pi(N, P) = 0.7$, $\pi(B, N) = 0.4$. Payoffs are $u_1(s_1, s_2) = \pi(s_1, s_2)$ and $u_2(s_1, s_2) = 1 - u_1(s_1, s_2) = \pi(s_2, s_1)$. Player 2's payoff function is the same rule applied with the roles swapped. It is the complement of player 1's because the probabilities of the two candidates winning add up to one: the game is constant-sum and symmetric.

(b)

$1 \setminus 2$	P	B	N
P	0.5, 0.5	0, 1	0.3, 0.7
B	1, 0	0.5, 0.5	0.4, 0.6
N	0.7, 0.3	0.6, 0.4	0.5, 0.5

(c) IESDS.

- Round 1. Player 1's rows over columns (P, B, N): $P = (0.5, 0, 0.3)$, $B = (1, 0.5, 0.4)$, $N = (0.7, 0.6, 0.5)$. P is strictly dominated by B ($0.5 < 1$, $0 < 0.5$, $0.3 < 0.4$) and also by N. B and N are not ranked (B is better against P, N against B and N). By symmetry the same holds for player 2. $S_i^1 = \{B, N\}$.
- Round 2. On $\{B, N\}$: $B = (0.5, 0.4)$, $N = (0.6, 0.5)$. B is strictly dominated by N. The same holds for player 2. $S_i^2 = \{N\}$.

$S^\infty = \{(N, N)\}$: both run negative campaigns and each wins with probability 0.5. The prediction is clear (a unique profile) after two rounds.

Interpretation. B survives round 1 only because it crushes a positive campaign. Once each candidate knows the other will not run a purely positive campaign, going fully negative wins. The outcome has the same win probabilities as (P, P), and in a constant-sum game every profile is Pareto optimal *for the candidates*. If voters dislike negative campaigning, the equilibrium is a prisoner's-dilemma outcome for them: the candidates end up exactly where they would have been with positive campaigns, but with a nastier campaign.

Verification. `check_A.py` builds the matrix from the four rules plus complementarity, checks symmetry, that P is dominated by both B and N, and that IESDS removes P (round 1) and B (round 2) for both players, leaving (N, N).

Answer. (a) $S_i = \{P, B, N\}$, $u_1 = \pi(s_1, s_2)$ from the rules, $u_2 = 1 - u_1$ (the same rule from candidate 2's side). (b) Matrix above. (c) Round 1 deletes P (dominated by B and N); round 2 deletes B (dominated by N); unique prediction (N, N), each wins with probability 0.5.

17 Tadelis Exercise 4.3

Exam · block B4

Statement (verbatim from the list, item 17)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.3 "Discrete First-Price Auction", p. 76 [PDF 95]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Discrete First-Price Auction:** An item is up for auction. Player 1 values the item at 3 while player 2 values the item at 5. Each player can bid either 0, 1, or 2. If player i bids more than player j then i wins the good and pays his bid, while the loser does not pay. If both players bid the same amount then a coin is tossed to determine who the winner is, and the winner gets the good and pays his bid while the loser pays nothing.
>
> a. Write down the game in matrix form.
>
> b. Does any player have a strictly dominated strategy?
>
> c. Which strategies survive IESDS?

What is asked. Discrete first-price auction: values $v_1 = 3$, $v_2 = 5$; bids in $\{0, 1, 2\}$; the higher bid wins and pays its bid, and ties are broken by a fair coin. (a) Write the matrix; (b) find strictly dominated strategies; (c) find the strategies that survive IESDS.

Method. Expected payoffs under the coin toss (payoffs are expected utilities, slide 18); dominance and IESDS (slides 23, 29).

Solution. **(a)** $u_1(b_1, b_2) = 3 - b_1$ if $b_1 > b_2$, 0 if $b_1 < b_2$, and $\frac{1}{2}(3 - b_1)$ if $b_1 = b_2$. The same rule with $v_2 = 5$ gives u_2.

$b_1 \backslash b_2$	0	1	2
0	1.5, 2.5	0, 4	0, 3
1	2, 0	1, 2	0, 3
2	1, 0	1, 0	0.5, 1.5

(b) Strictly dominated strategies (full game).

- Player 1, over $b_2 = (0, 1, 2)$: $0 \mapsto (1.5, 0, 0)$, $1 \mapsto (2, 1, 0)$, $2 \mapsto (1, 1, 0.5)$. Bid 0 is beaten by bid 1 against $b_2 = 0, 1$ but ties at $b_2 = 2$: only weakly dominated. Bids 1 and 2 are not ranked: 1 beats 2 against $b_2 = 0$ $(2 > 1)$, 2 beats 1 against $b_2 = 2$ $(0.5 > 0)$. Bid 0 is not dominated by 2 either $(1.5 > 1$ against $b_2 = 0)$. *Player 1 has no strictly dominated strategy.*

- Player 2, over $b_1 = (0, 1, 2)$: $0 \mapsto (2.5, 0, 0)$, $1 \mapsto (4, 2, 0)$, $2 \mapsto (3, 3, 1.5)$. Bid 0 is strictly dominated by bid 2 ($2.5 < 3$, $0 < 3$, $0 < 1.5$); it is only weakly dominated by bid 1. Bids 1 and 2 are not ranked. *Player 2's bid* 0 *is strictly dominated (by* 2*).*

(c) IESDS.

1. Round 1: delete $b_2 = 0$. $S_1^1 = \{0, 1, 2\}$, $S_2^1 = \{1, 2\}$.
2. Round 2: player 1 over $b_2 \in \{1, 2\}$: $0 \mapsto (0, 0)$, $1 \mapsto (1, 0)$, $2 \mapsto (1, 0.5)$. Bid 0 is strictly dominated by bid 2. Bid 1 is only weakly dominated by 2. $S_1^2 = \{1, 2\}$.
3. Round 3: player 2 over $b_1 \in \{1, 2\}$: $1 \mapsto (2, 0)$, $2 \mapsto (3, 1.5)$. Bid 1 is strictly dominated by 2. $S_2^3 = \{2\}$.
4. Round 4: player 1 against $b_2 = 2$: bid 1 gives 0, bid 2 gives 0.5. Delete 1. $S_1^4 = \{2\}$.

$S^\infty = \{(2, 2)\}$, with expected payoffs $(0.5, 1.5)$: both bid the maximum, and each wins with probability $\frac{1}{2}$.

Interpretation. Four rounds means four layers of reasoning: player 2 is rational; player 1 knows it; player 2 knows that player 1 knows it; and so on. The discreteness matters: with only three bids, the high-value bidder cannot outbid at 2, so the auction ends in a coin toss at the top bid. Bidder 2 would like to bid 3 and win for sure. In a first-price auction bidders shade below their values (player 1 bids $2 < 3$); contrast this with item 23, where bidding one's value is weakly dominant.

Verification. `check_A.py` builds the matrix from the rules (exact fractions), checks every dominance claim in (b), and runs IESDS: removals are $b_2 = 0$, then $b_1 = 0$, then $b_2 = 1$, then $b_1 = 1$, leaving $(2, 2)$.

Answer. (a) Matrix above. (b) Only player 2's bid 0 is strictly dominated (by 2); player 1 has none (his 0 is only weakly dominated). (c) IESDS removes $b_2 = 0$, $b_1 = 0$, $b_2 = 1$, $b_1 = 1$ in turn; the unique survivor is $(2, 2)$ with expected payoffs $(0.5, 1.5)$.

18 Tadelis Exercise 4.10

Exam · block B4

Statement (verbatim from the list, item 18)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.10 "Popsicle Stands", p. 78 [PDF 97]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Popsicle Stands:** Five lifeguard towers are lined up along a beach; the leftmost tower is number 1 and the rightmost tower is number 5. Two vendors, players 1 and 2, each have a popsicle stand that can be located next to one of five towers. There are 25 people located next to each tower, and each person will purchase a popsicle from the stand that is closest to him or her. That is, if player 1 locates his stand at tower 2 and player 2 at tower 3, then 50 people (at towers 1 and 2) will purchase from player 1, while 75 (from towers 3, 4, and 5) will purchase from vendor 2. Each purchase yields a profit of \$1.
>
> a. Specify the strategy set for each player. Are there any strictly dominated strategies?
>
> b. Find the set of strategies that survive rationalizability.

What is asked. Two popsicle vendors each place a stand at one of five equally spaced towers, each with 25 customers who buy from the nearest stand. (a) Give the strategy sets and any strictly dominated strategies; (b) find the strategies that survive rationalizability.

Method. Normal form and IESDS (slides 29–33, the spatial-competition application); for (b), rationalizability (Tadelis §4.3, pp. 69–75, covered in Lecture 2) and the fact that it can only refine IESDS. As in the slides' mini case, a customer who is equally distant from both stands splits evenly, and two stands at the same tower split all customers.

Solution. (a) Strategy sets and payoffs. $I = \{1, 2\}$, $S_1 = S_2 = \{1, 2, 3, 4, 5\}$. If $s_1 < s_2$, vendor 1 serves towers $1, \ldots, s_1$ and half of the $s_2 - s_1 - 1$ towers between the stands (those nearer to him, plus half of a middle tower if there is one):

$$u_1(s_1, s_2) = 25\Big(s_1 + \frac{s_2 - s_1 - 1}{2}\Big) = \frac{25\,(s_1 + s_2 - 1)}{2}, \qquad u_1(s, s) = 62.5,$$

and $u_1(s_1, s_2) = 125 - u_2(s_1, s_2)$, with the game symmetric. (Check: $u_1(2, 3) = 50$, the book's example.) Vendor 1's payoffs, rows s_1, columns s_2:

$s_1 \backslash s_2$	1	2	3	4	5
1	62.5	25	37.5	50	62.5
2	100	62.5	50	62.5	75
3	87.5	75	62.5	75	87.5
4	75	62.5	50	62.5	100
5	62.5	50	37.5	25	62.5

Strictly dominated strategies: tower 1 is strictly dominated by tower 2, row by row $62.5 < 100$, $25 < 62.5$, $37.5 < 50$, $50 < 62.5$, $62.5 < 75$ (it is also dominated by 3 and 4). By symmetry, tower 5 is strictly dominated by tower 4. Towers 2, 3, 4 are not dominated: 2 is the best reply to 1, 4 to 5, 3 to 3. The same holds for vendor 2.

IESDS. Round 1: $S_i^1 = \{2, 3, 4\}$. Round 2: on columns $\{2, 3, 4\}$, row 2 gives $(62.5, 50, 62.5)$, row 3 gives $(75, 62.5, 75)$, row 4 gives $(62.5, 50, 62.5)$: towers 2 and 4 are strictly dominated by 3. $S_i^2 = \{3\}$, so $S^\infty = \{(3, 3)\}$ with payoffs $(62.5, 62.5)$.

(b) Rationalizability. Let R_i^k be the strategies left after k rounds of deleting strategies that are never a best response (to beliefs over the opponent's remaining strategies).

- $R^k \subseteq S^k$ for all k, by induction. If $R^k \subseteq S^k$ and s_i is strictly dominated given S_{-i}^k, it is strictly dominated given the smaller set R_{-i}^k, so it is never a best response there and is deleted. Hence rationalizable strategies survive IESDS: $R_i^\infty \subseteq \{3\}$.
- Tower 3 is a best reply to tower 3 (62.5 vs. 50 for towers 2 and 4, 37.5 for 1 and 5), so the pair $(3, 3)$ justifies itself in every round and is never deleted.

The rationalizable set is $\{3\}$ for each vendor: both stands at the middle tower.

Interpretation. This is Hotelling's principle of minimum differentiation: each vendor moves toward the center to capture more of the other's half, until both cluster. Clustering is bad for customers. Total walking at $(3, 3)$ is $25(2 + 1 + 0 + 1 + 2) = 150$ tower-lengths, against $25(1 + 0 + 1 + 0 + 1) = 75$ at $(2, 4)$, which also splits the market 62.5/62.5. Same logic as the pharmacy mini case of slides 31–33 (graded in class, so not solved here).

Verification. `check_A.py` computes the payoffs customer by customer, confirms the book's example and the closed form $25(s_1 + s_2 - 1)/2$, the dominance of tower 1 by 2, 3 and 4, and the IESDS rounds ($\{1, 5\}$, then $\{2, 4\}$), leaving $(3, 3)$.

Answer. (a) $S_i = \{1, \ldots, 5\}$; tower 1 is strictly dominated by tower 2 and tower 5 by tower 4. (b) After deleting 1 and 5, towers 2 and 4 are strictly dominated by 3. Since rationalizable strategies survive IESDS and $(3, 3)$ is self-justifying, only tower 3 is rationalizable for each vendor.

Tier 3 — Advanced

19 Muñoz-Garcia Ch. 1, Exercise 16

Advanced · block B3

Statement (verbatim from the list, item 19)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 1, Exercise 16 "Equilibrium Predictions from IDSDS Versus IDWDS", p. 31 [PDF 44]. Reproduced verbatim (statement only; the book's worked answer follows it).

> In previous exercises applying IDSDS we sometimes started finding strictly dominated strategies for the row player, while in other exercises we began identifying strictly dominated strategies for the column player (or the matrix player in games with three players). While the order of deletion does not affect the equilibrium outcome when applying IDSDS, it *can* affect the set of equilibrium outcomes when we delete weakly (rather than strictly) dominated strategies.
>
> Use the game in Fig. 1.37 to show that the order in which weakly dominated strategies are eliminated can affect equilibrium outcomes.

Player 1 Player 2	L	M	R
U	2, 1	1, 1	0, 0
C	1, 2	3, 1	2, 1
D	2, -2	1, -1	-1, -1

Fig. 1.37 Normal-form game

What is asked. Use the 3×3 game of Fig. 1.37 to show that the order in which *weakly* dominated strategies are eliminated (IDWDS) can change the result, while the order does not matter under strict dominance.

Method. Weak dominance (slide 23); IESDS (slide 29); order independence of strict deletion (MWG §8.B, pp. 239–240, and Exercise 8.B.4, item 20); MWG's example of order-dependent weak deletion (p. 240).

Solution. Matrix from the book (p. 31):

$1 \setminus 2$	L	M	R
U	2, 1	1, 1	0, 0
C	1, 2	3, 1	2, 1
D	2, −2	1, −1	−1, −1

Strict dominance first. Player 1: D ties U against L and M; C loses to U and D against L; U and D lose to C against M. Player 2, columns over rows (U, C, D): $L = (1, 2, -2)$, $M = (1, 1, -1)$, $R = (0, 1, -1)$; R ties M in rows C and D, and R beats L in row D. *No strategy is strictly dominated*, so IESDS keeps all nine profiles, whatever the order. The weak relations are: U weakly dominates D ($2 = 2$, $1 = 1$, $0 > -1$), and M weakly dominates R ($1 > 0$, $1 = 1$, $-1 = -1$).

Route 1 (start with player 1).

1. Delete D (weakly dominated by U).
2. On rows $\{U, C\}$: $L = (1, 2)$ beats $R = (0, 1)$ *strictly*. Delete R.
3. On rows $\{U, C\}$: $L = (1, 2)$ vs $M = (1, 1)$: M is weakly dominated by L. Delete M.
4. Against L only: U gives 2, C gives 1. Delete C.

Result: the single profile (U, L), payoffs $(2, 1)$.

Route 2 (start with player 2).

1. Delete R (weakly dominated by M).
2. On columns $\{L, M\}$: $U = (2, 1)$ and $D = (2, 1)$ are identical, so neither weakly dominates the other; $C = (1, 3)$ is not ranked against them. Player 2 on rows $\{U, C, D\}$: $L = (1, 2, -2)$, $M = (1, 1, -1)$, not ranked. Nothing more can be deleted.

Result: $\{U, C, D\} \times \{L, M\}$, six profiles.

The two orders give different predictions, $\{(U, L)\}$ versus six profiles. Deleting *all* weakly dominated strategies of both players at once in each round (D and R together, then M, then C) also ends at (U, L); the six-profile result comes from deleting R alone first.

Why weak deletion is order-dependent and strict deletion is not. Strict dominance survives shrinking the opponent's set: if s_i' beats s_i against every $s_{-i} \in S_{-i}^k$, it still does against any subset. So a strategy that can be deleted now can still be deleted later, and the end result does not depend on the order (proved in item 20). Weak dominance can be *destroyed* by deleting the opponent's strategies. U weakly dominates D only because of column R ($0 > -1$). Once R is deleted first, U and D tie everywhere and D can no longer be removed. Its presence then keeps M alive for player 2 ($-1 > -2$ in row D).

Interpretation. Weak deletion is a tempting shortcut, but its predictions depend on an arbitrary procedural choice, which is why the lecture's solution concept uses strict dominance. The common-knowledge justification also fails: deleting D because of column R, and then deleting R, is self-undermining.

Verification. `check_A.py` confirms that IESDS deletes nothing, replays both routes asserting that each deletion is a valid weak (or strict) dominance at that stage, confirms that nothing is weakly dominated at the end of route 2, and that simultaneous weak deletion ends at (U, L). `check_C.py` runs 400 random weak-deletion orders and finds both end results. The book's answer (pp. 31–33) agrees: route 1 gives (U, L) and route 2 gives the six profiles.

Answer. No strategy is strictly dominated, so IESDS keeps all nine profiles in any order. Weakly: deleting D, then R (now strictly dominated), M, C gives (U, L). Deleting R first leaves U and D tied and stops at $\{U, C, D\} \times \{L, M\}$. The order matters for weak deletion because weak dominance can disappear when the opponent's strategies are removed.

20 MWG Exercise 8.B.4

Advanced · block B3

Statement (verbatim from the list, item 20)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.B.4ˆC, p. 262 [PDF 142]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

> Show that the order of deletion does not matter for the set of strategies surviving a process of iterated deletion of strictly dominated strategies.

What is asked. Show that the order of deletion does not matter for the set of strategies that survive iterated deletion of strictly dominated strategies.

Method. Proof for finite games with dominance by pure strategies (MWG §8.B, pp. 239–240; Lecture 1, slide 29). Two facts do the work: strict dominance is inherited by smaller opponent sets, and in a finite game a strictly dominated strategy has an undominated dominator.

Solution. **Setup.** $G = (I, \{S_i\}, \{u_i\})$ with every S_i finite. For a product set $T = \prod_i T_i \subseteq S$, say $s_i \in T_i$ is *dominated in* T if some $t_i \in T_i$ satisfies $u_i(t_i, s_{-i}) > u_i(s_i, s_{-i})$ for all $s_{-i} \in T_{-i}$. A *deletion sequence* is $S = T^0 \supsetneq T^1 \supsetneq \cdots \supsetneq T^K$, where each step removes a nonempty set of strategies that are dominated in the current set (any players, any number, in any order), and where nothing is dominated in T^K. The lecture's procedure (remove everything dominated each round) is one such sequence. *Claim:* every deletion sequence ends at the same T^K.

Two facts.

(i) *Inheritance.* If t_i beats s_i against every $s_{-i} \in A_{-i}$ and $B_{-i} \subseteq A_{-i}$, it beats s_i against every $s_{-i} \in B_{-i}$.

(ii) *Transitivity.* If t_i beats s_i and r_i beats t_i on the same set, then r_i beats s_i there.

Proof. Let $(A^k)_{k=0}^K$ and $(B^m)_{m=0}^M$ be two deletion sequences with end sets $A = A^K$ and $B = B^M$. We show $B \subseteq A^k$ for all k, by induction. It holds for $k = 0$. Suppose $B \subseteq A^k$ and that some $s_i \in B_i$ is deleted at step $k+1$ of the A-sequence. Then some $t_i \in A_i^k$ beats s_i on $A_{-i}^k \supseteq B_{-i}$, so by (i) the set

$$D = \{\, r_i \in S_i : u_i(r_i, s_{-i}) > u_i(s_i, s_{-i}) \ \forall s_{-i} \in B_{-i} \,\}$$

is nonempty. We show that $D \cap B_i \neq \emptyset$. Then s_i is dominated in B, which contradicts the fact that the B-sequence stopped.

Suppose instead that every $r_i \in D$ is deleted at some step of the B-sequence. Let m be the last step at which an element of D is deleted (D is finite), and let D_m be the elements of D deleted at step m. "Beats on B_{-i}^{m-1}" is a strict partial order on D_m (irreflexive and transitive), so D_m has

an element r^* that no other element of D_m beats on B_{-i}^{m-1}. Since r^* is deleted at step m, some $q \in B_i^{m-1}$ beats r^* on $B_{-i}^{m-1} \supseteq B_{-i}$. By (i) and (ii), q beats s_i on B_{-i}, so $q \in D$. Now $q \in B_i^{m-1}$ and $q \in D$, so either q is deleted at step m, and then $q \in D_m$ beats r^*, contradicting the choice of r^*; or q is deleted after m, contradicting the choice of m; or q is never deleted, so $q \in D \cap B_i$, contradicting our supposition. Hence $D \cap B_i \neq \emptyset$.

So no strategy of B is ever deleted by the A-sequence: $B \subseteq A$. Exchanging the roles of the two sequences gives $A \subseteq B$, so $A = B$. □

Remarks.

- The selection of r^* handles a subtle point: the strategy that dominates s_i may itself be deleted, possibly in the same step. Finiteness guarantees that the chain of dominators ends at a strategy that survives.
- Finiteness matters. In infinite games a strategy can be dominated only by strategies that are themselves dominated, and different orders can then end at different sets. Dufwenberg and Stegeman (2002, *Econometrica*) give such examples. In the continuous games of items 15 and 21 there is no such problem, because each round's survivors are exactly the possible best replies.
- The argument breaks for *weak* dominance at step (i): weak dominance on A_{-i} need not survive on a subset B_{-i}, because the strict inequality may sit on a deleted s_{-i} (item 19).

Interpretation. The set S^∞ is a property of the game, not of the analyst's bookkeeping. One may delete in whatever order is convenient (one player at a time, one strategy at a time) and the prediction is the same. This is part of what makes IESDS a well-defined solution concept.

Verification. `check_C.py` draws 3 000 random finite games (between 2×2 and 5×5, payoffs 0–6). For each it runs the lecture's simultaneous procedure and five random sequences that each delete a random nonempty subset of one random player's dominated strategies. All end at the same set. *Comparison with the MWG solutions manual* (pp. 8-2–8-3): it shows that a strategy strictly dominated at some round will be deleted at the next round, assuming that its dominator s_i^* is "not a strictly dominated strategy". It does not prove that such an undominated dominator exists (the finiteness step above), and it does not complete the comparison of two whole sequences. The conclusion is the same; the proof above fills those two gaps. The manual's note on weak dominance agrees with item 19.

Answer. In a finite game every deletion sequence ends at the same set. A strategy in one end set B can never be deleted by another sequence. If it were, a dominator chain, cut off at a maximal element of the last-deleted batch, would produce a strategy in B that dominates it on B_{-i}, contradicting that B is final. Strict dominance is inherited by smaller opponent sets; weak dominance is not, which is why weak deletion is order-dependent.

21 MWG Exercise 8.B.5

Advanced · block B3

Statement (verbatim from the list, item 21)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.B.5ˆC, p. 262 [PDF 142]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

Consider the Cournot duopoly model (discussed extensively in Chapter 12) in which two firms, 1 and 2, simultaneously choose the quantities they will sell on the market, q_1 and q_2. The price each receives for each unit given these quantities is $P(q_1, q_2) = a - b(q_1 + q_2)$. Their costs are c per unit sold.

(a) Argue that successive elimination of strictly dominated strategies yields a unique prediction in this game.

(b) Would this be true if there were three firms instead of two?

What is asked. Linear Cournot with price $a - b(q_1 + q_2)$ and unit cost c: (a) show that iterated deletion of strictly dominated strategies gives a unique prediction; (b) say whether this remains true with three firms.

Method. IESDS by bounding best replies, as in Tadelis's Cournot example (§4.2.2, pp. 65–67) and item 15. Assume $a > c \geq 0$, $b > 0$, quantities $q_i \in [0, \infty)$, and write $m = (a - c)/(2b)$ for the monopoly output.

Solution. Payoff and best reply. With Q_{-i} the others' total output,

$$u_i = \big(a - b(q_i + Q_{-i}) - c\big)\, q_i = b\, q_i\, (2x - q_i), \qquad x = m - \tfrac{Q_{-i}}{2},$$

a concave parabola in q_i with peak at x. Hence $BR_i(Q_{-i}) = \max\{0,\, m - Q_{-i}/2\}$, and, as in item 15,

$$u_i(a', Q_{-i}) - u_i(t, Q_{-i}) = b\,(t - a')(t + a' - 2x). \tag{1}$$

So if every possible peak x lies in $[\underline{a}, \bar{a}]$, then outputs above $\bar{a}$ are strictly dominated by $\bar{a}$ and outputs below $\underline{a}$ by $\underline{a}$. An output that is the unique best reply to some surviving Q_{-i} is not dominated.

(a) Two firms. Here $Q_{-i} = q_j$. If $S_j^k = [L_k, U_k]$, the surviving best replies fill $[BR(U_k), BR(L_k)]$, so

$$L_{k+1} = \max\{0,\ m - \tfrac{U_k}{2}\}, \qquad U_{k+1} = m - \tfrac{L_k}{2}.$$

- Round 1: $q_j \in [0, \infty)$, so the best replies fill $[0, m]$ ($BR = m$ at $q_j = 0$, $BR = 0$ for $q_j \geq 2m$). Every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: $q_j \leq m$ gives $BR \geq m/2$. $S_i^2 = [m/2, m]$.
- Round 3: $S_i^3 = [m/2, 3m/4]$; round 4: $[5m/8, 3m/4]$; …

From round 2 on, the bounds are interior and $U_{k+1} - L_{k+1} = \frac{1}{2}(U_k - L_k)$, so the width is $m/2^{k-1} \to 0$. Any limit point solves $q = m - q/2$, that is,

$$S_i^\infty = \left\{\tfrac{2m}{3}\right\} = \left\{\frac{a - c}{3b}\right\}, \qquad i = 1, 2.$$

IESDS predicts the Cournot outcome $q_1 = q_2 = (a - c)/(3b)$. (Tadelis's example, with $m = 45$, gives 30 in the same way.)

(b) Three firms. Now $Q_{-i} = q_j + q_h$.

- Round 1: $Q_{-i} \in [0, \infty)$, so the best replies fill $[0, m]$ and every $q_i > m$ is strictly dominated by m. $S_i^1 = [0, m]$.
- Round 2: now $Q_{-i} \in [0, 2m]$, and $BR(2m) = \max\{0, m - m\} = 0$. The best replies still fill all of $[0, m]$: each $q \in [0, m]$ is the unique best reply to $q_j = q_h = m - q \in [0, m]$, which survived round 1. Nothing is deleted: $S_i^2 = S_i^1$.

The process stops after one round with $S_i^\infty = [0, (a-c)/(2b)]$ for each firm. *No*, IESDS does not give a unique prediction with three firms. (The Cournot equilibrium $q_i = (a-c)/(4b)$ lies inside, but so does everything else in $[0, m]$.)

Interpretation. With two firms the best replies form a contraction: each round of "I know that you know ..." halves the range of plausible outputs. With three firms the rivals' *combined* output can be as large as $2m$, enough to make zero output a best reply, so the first-round bounds reproduce themselves and higher-order reasoning adds nothing. IESDS is weak in games with several players whose actions are strategic substitutes; Nash equilibrium (Lecture 2) still pins down the outcome.

Verification. `check_B.py` verifies with sympy that the Cournot payoff equals $bq_i(2x - q_i)$ with $x = m - q_j/2$, the factorization (1), and that the fixed point of $q = m - q/2$ is $(a-c)/(3b)$. It runs the duopoly interval recursion for 60 rounds in exact arithmetic ($m = 1$: $[0,1]$, $[\frac{1}{2}, 1]$, $[\frac{1}{2}, \frac{3}{4}]$, ..., $\to \frac{2}{3}$), checks that every $q \in [0, m]$ is the best reply to $Q = 2(m-q)$ with both rivals in $[0, m]$, and runs brute-force IESDS for three firms on a grid ($a - c = 12$, $b = 1$, steps of 0.5), which leaves exactly $[0, 6] = [0, m]$. *Comparison with the MWG solutions manual* (pp. 8-3–8-4): it agrees on both parts, $(a-c)/(3b)$ for two firms and $q_i \in [0, (a-c)/(2b)]$ with no unique prediction for three.

Answer. (a) $BR_i(q_j) = \max\{0, (a-c)/(2b) - q_j/2\}$; IESDS gives $[0, m]$, $[m/2, m]$, $[m/2, 3m/4]$, ..., with widths halving, so the unique survivor is $q_1 = q_2 = (a-c)/(3b)$. (b) No: with three firms the rivals' total can reach $2m$, every $q_i \in [0, (a-c)/(2b)]$ is a best reply to some surviving profile, and IESDS stops at $[0, (a-c)/(2b)]$ for each firm.

22 MWG Exercise 8.B.7

Advanced · block B2

Statement (verbatim from the list, item 22)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.B.7ˆB, p. 262 [PDF 142]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

> Show that any strictly dominant strategy in game $[I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$ must be a pure strategy.

What is asked. Show that a strictly dominant strategy of the mixed extension of a game, $[I, \{\Delta(S_i)\}, \{u_i(\cdot)\}]$, must be a pure strategy.

Method. Mixed strategies and dominance in the mixed extension (MWG Def. 8.B.4, p. 240; Prop. 8.B.1 and Exercise 8.B.6, p. 241). Key fact: expected utility is linear in one's own mixing probabilities, so a mixture's payoff is a weighted average of the payoffs of the pure strategies it uses. Mixed strategies are the subject of Lecture 2; only this averaging property is needed here.

Solution. Fix player i and a mixed strategy $\sigma_i \in \Delta(S_i)$. σ_i is *strictly dominant* if $u_i(\sigma_i, \sigma_{-i}) > u_i(\sigma_i', \sigma_{-i})$ for every $\sigma_i' \neq \sigma_i$ and every σ_{-i}. In particular this must hold against every pure profile s_{-i} (a degenerate σ_{-i}) and every pure $\sigma_i' = s_i \neq \sigma_i$.

Proof. Suppose σ_i is strictly dominant but not pure: its support contains at least two pure strategies $s_i^1, \ldots, s_i^N$ ($N \geq 2$), with $\sigma_i(s_i^n) > 0$ and $\sum_n \sigma_i(s_i^n) = 1$. Each s_i^n is then a strategy different from σ_i. Fix any $s_{-i} \in S_{-i}$. By linearity of expected utility,

$$u_i(\sigma_i, s_{-i}) = \sum_{n=1}^{N} \sigma_i(s_i^n)\, u_i(s_i^n, s_{-i}) \leq \max_n u_i(s_i^n, s_{-i}).$$

Let n^* attain the maximum. Then $u_i(\sigma_i, s_{-i}) \leq u_i(s_i^{n^*}, s_{-i})$, while strict dominance of σ_i requires $u_i(\sigma_i, s_{-i}) > u_i(s_i^{n^*}, s_{-i})$. Contradiction. Hence the support of σ_i is a single pure strategy: σ_i is pure. □

Remarks.

- The argument needs only *one* opponent profile s_{-i}; it uses no finiteness of S_{-i} and works for any I.
- It fails for *weak* dominance in the "$\geq$" sense. If two pure strategies tie against everything (as U and D in item 12), every mixture of them is also "$\geq$-dominant".
- The converse direction is different: a pure strategy can be strictly *dominated* by a mixed strategy without being dominated by any pure one (MWG Fig. 8.B.5, pp. 240–241). Mixing can help in eliminating strategies, never in creating a dominant one.

Interpretation. A strictly dominant strategy must be the unique best action against every opponent profile. A mixture is never strictly better than the best of the pure strategies it mixes, so mixing can only be optimal when the player is indifferent among those strategies, and indifference is incompatible with strict dominance. This is the same reason why, in Lecture 2, a player mixes only among strategies that give her equal payoffs.

Verification. `check_C.py` draws 20 000 random supports (2–4 pure strategies, random payoffs and weights, exact fractions) and confirms that the mixture's payoff never exceeds the best support payoff, so it never strictly beats every pure strategy in its support. *Comparison with the MWG solutions manual* (p. 8-4): same argument (a strict inequality against each pure strategy in the support, averaged with the mixing weights, contradicts linearity). It agrees.

Answer. If σ_i mixes over $N \geq 2$ pure strategies, then for any s_{-i} its payoff is a weighted average of theirs, so it is at most $\max_n u_i(s_i^n, s_{-i})$. It cannot strictly beat that best pure strategy, which contradicts strict dominance. Hence every strictly dominant strategy is pure.

23 MWG Exercise 8.B.3

Advanced · block B4

Statement (verbatim from the list, item 23)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 8 Exercises, Exercise 8.B.3^B, p. 262 [PDF 142]. Reproduced verbatim. Answer in the source: none in the book (see the MWG solutions manual in `Livros/`).

> Consider the following auction (known as a *second-price*, or *Vickrey*, auction). An object is auctioned off to I bidders. Bidder i's valuation of the object (in monetary terms) is v_i. The auction rules are that each player submit a bid (a nonnegative number) in a sealed envelope. The envelopes are then opened, and the bidder who has submitted the highest bid gets the object but pays the auctioneer the amount of the *second-highest* bid. If more than one bidder

submits the highest bid, each gets the object with equal probability. Show that submitting a bid of v_i with certainty is a weakly dominant strategy for bidder i. Also argue that this is bidder i's unique weakly dominant strategy.

What is asked. In a sealed-bid second-price (Vickrey) auction with I bidders and private valuations v_i (ties broken at random), show that bidding v_i is weakly dominant for bidder i, and that it is bidder i's unique weakly dominant strategy.

Method. Weak dominance (MWG Def. 8.B.3, pp. 237–238; Lecture 1, slide 23) by case analysis on the highest competing bid; the second-price auction is one of the slides' opening examples (slide 4). We use the lecture's definition: b weakly dominates b' if it does at least as well against every opponent profile and strictly better against at least one.

Solution. Fix bidder i ($I \geq 2$) and let $h = \max_{j \neq i} b_j \geq 0$ be the highest rival bid and $k \geq 1$ the number of rivals bidding h. Bidder i's payoff from bid b depends on the rivals only through (h, k):

$$u_i(b; h, k) = \begin{cases} v_i - h & b > h, \\ \dfrac{v_i - h}{k+1} & b = h, \\ 0 & b < h. \end{cases}$$

The price never depends on bidder i's own bid; the bid only decides *whether* he wins.

Step 1: v_i is a best reply to every (h, k). If $h < v_i$, bidding v_i wins and earns $v_i - h > 0$, the most any bid can earn. If $h > v_i$, bidding v_i loses and earns 0, while any winning or tying bid earns $v_i - h < 0$ or a share of it. If $h = v_i$, every bid earns 0. So $u_i(v_i; h, k) \geq u_i(b; h, k)$ for every bid b and every rival profile.

Step 2: strict somewhere against each alternative.

- $b > v_i$ (overbidding): take $h \in (v_i, b)$. Bid b wins and pays $h > v_i$, earning $v_i - h < 0$; bid v_i loses and earns 0.
- $b < v_i$ (underbidding): take $h \in (b, v_i)$. Bid b loses and earns 0; bid v_i wins and earns $v_i - h > 0$.

With Step 1, v_i weakly dominates every other bid: *bidding v_i is weakly dominant.*

Step 3: uniqueness. Let $b \neq v_i$. The rival profile constructed in Step 2 gives $u_i(b; h, \cdot) < u_i(v_i; h, \cdot)$, so b is not even a best reply there, and a fortiori not weakly dominant. Mixed bids (MWG's "with certainty" refers to them) fail too. If σ puts probability $p > 0$ on bids below v_i, there is an $h < v_i$ with $\sigma([0, h]) > 0$ (since $[0, v_i) = \bigcup_n [0, v_i - 1/n]$). Against that h, every bid $b \leq h$ earns strictly less than v_i does (0 or $(v_i - h)/(k+1) < v_i - h$), and no bid earns more. So σ is strictly worse than v_i there. The case of mass above v_i is symmetric. Hence bidding v_i with certainty is the *unique* weakly dominant strategy.

Interpretation. Truth-telling is optimal whatever the others do, and whatever bidder i believes about them: no knowledge of rivals' values, rationality or bids is needed, which is much less than IESDS or Nash equilibrium require. The second-price rule separates *whether* you win (your bid) from *what* you pay (others' bids), so the only effect of shading or inflating a bid is to win in the wrong cases. In a first-price auction the bid also sets the price, and bidders shade (item 17). Truthful bidding by all is a weakly dominant-strategy equilibrium and allocates the object efficiently, to the highest valuation.

Verification. `check_C.py` enumerates bids on the grid $\{0, \dots, 8\}$ with one and with two rivals and values $v \in \{0, 3, 8\}$. In every case v weakly dominates each other grid bid, and v is the only grid bid that does at least as well as every other bid against every rival profile. *Comparison with the MWG solutions manual* (p. 8-2): it gives the same two-case argument (overbidding and underbidding are weakly dominated by v_i) but does not argue the "unique" part, which Step 3 supplies. It does not treat ties explicitly either; here they are handled by the $(v_i - h)/(k+1)$ term.

Answer. The price paid is the highest rival bid h, independent of one's own bid. Bidding v_i wins exactly when $h < v_i$, that is, exactly when winning is profitable, so it is a best reply to every rival profile. Any $b > v_i$ does strictly worse when $h \in (v_i, b)$, and any $b < v_i$ when $h \in (b, v_i)$. Hence bidding v_i is weakly dominant, and no other bid, pure or mixed, is.

24 Tadelis Exercise 4.4

Advanced · block B4

Statement (verbatim from the list, item 24)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 4.4 "eBay's Recommendation", pp. 76–77 [PDF 95–96]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **eBay's Recommendation:** It is hard to imagine that anyone is not familiar with eBay, the most popular auction web site by far. In a typical eBay auction a good is placed for sale, and each bidder places a "proxy bid," which eBay keeps in memory. If you enter a proxy bid that is lower than the current highest bid, then your bid is ignored. If, however, it is higher, then the current bid increases up to one increment (say, \$0.01) above the *second highest* proxy bid. For example, imagine that three people have placed bids on a used laptop of \$55, \$98, and \$112. The current price will be at \$98.01, and if the auction ended the player who bid \$112 would win at a price of \$98.01. If you were to place a bid of \$103.45 then the player who bid \$112 would still win, but at a price of \$103.46, while if your bid was \$123.12 then you would win at a price of \$112.01.
>
> Now consider eBay's historical recommendation that you think hard about the value you impute to the good and that you enter your true value as your bid—no more, no less. Assume that the value of the good for each potential bidder is independent of how much other bidders value it.
>
> a. Argue that bidding more than your valuation is weakly dominated by actually bidding your valuation.
> b. Argue that bidding less than your valuation is weakly dominated by actually bidding your valuation.
> c. Use your analysis to make sense of eBay's recommendation. Would you follow it?

Optional extras.

- Muñoz-Garcia Ch. 1 Ex. 6–8 (pp. 11–14) and Ex. 11 (p. 20, three players): more IESDS drill, all with answers.
- Muñoz-Garcia Ch. 1 Ex. 17 (p. 33): a second weak-deletion order example, after item 19.
- Tadelis 3.1, 3.3, 3.5 (p. 58): the other verbal-to-matrix stories, if items 1–2 felt slow.

What is asked. Under eBay's proxy bidding (the winner pays one increment above the second-highest proxy bid) with independent private values: (a) argue that bidding above one's value is weakly dominated by bidding one's value; (b) the same for bidding below; (c) make sense of eBay's advice to bid one's true value, and say whether you would follow it.

Method. Weak dominance (Lecture 1, slide 23; Tadelis Remark, p. 63), by comparing outcomes case by case in the highest rival proxy bid. As the increment (\$0.01) goes to zero, proxy bidding *is* the second-price auction of item 23. We first argue in that limit, which is the exercise's intent, and then check what the increment changes.

Solution. Let v be your value, b your proxy bid, h the highest rival proxy bid. In the limit model you win iff $b > h$ (ties split at random) and pay h, whatever b is.

(a) Overbidding, $b > v$. Compare with bidding v:

- $h < v$: both bids win at price h; same payoff $v - h$.
- $h > b$: both lose; payoff 0 for both.
- $v \leq h \leq b$: bid v earns 0 (it loses, or ties at a price equal to v), while bid b wins (or ties) at price $h \geq v$, earning $v - h \leq 0$. For $v < h < b$ this is strictly negative.

Bidding v is never worse and is strictly better when $v < h < b$: b is weakly dominated by v. Overbidding only matters when it makes you win at a price above your value.

(b) Underbidding, $b < v$.

- $h < b$: both win at price h; same payoff.
- $h > v$: both lose; same payoff 0.
- $b \leq h < v$: bid v wins and earns $v - h > 0$; bid b loses (or ties) and earns 0 (or $(v - h)/2$). Strictly worse.
- $h = v$: both earn 0.

So b is weakly dominated by v. Underbidding only matters when it makes you lose an auction you would have won profitably; it never lowers the price you pay.

What the increment changes. With a positive increment the price is $\min\{b,\ h + \text{increment}\}$ (the winner never pays above his own proxy). `check_C.py` enumerates this rule with bids in cents, a one-cent increment, two bidders, random tie-breaking and $v = \$10.00$.

- (a) holds exactly: v weakly dominates every $b > v$.
- (b) holds for every $b \leq v - 3$ cents. Bidding $v - 2$ cents is payoff-equivalent to bidding v. Bidding $v - 1$ cent *weakly dominates* bidding v: it ties with probability $\frac{1}{2}$ when $h = v - 1$ cent (expected gain half a cent), where bidding v wins at a price of exactly v and earns nothing.

So the exact statement in (b) holds only up to the size of one increment. Any gain from shading is below one cent, and it disappears in the limit model the exercise intends.

(c) eBay's recommendation. By (a) and (b), bidding your true value is (weakly) optimal whatever the other bidders do: you do not need to guess their values, their strategies, or how many of them there are. Proxy bidding turns eBay's ascending auction into a second-price auction, and the advice "enter your true value, no more, no less" is exactly the weakly dominant strategy of item 23. I would follow it, with caveats that come from features the model leaves out:

- *Dynamics.* Real eBay auctions have a fixed end time and bids arrive over time. Late bidding ("sniping") is common because it gives rivals no time to react, for example to incremental bidders who raise their bids step by step. If everyone bids their value once, the timing does not matter.
- *Values not independent.* For goods with resale or common value, rivals' bids carry information about the value (the winner's curse), and "your value" must take that into account.
- *Increments and shill bidding.* Shading by a few cents is harmless (above); a seller bidding on her own item to raise h does not change the advice either, because your bid still never affects your price.

Interpretation. The general lesson: when your bid affects only *whether* you win and not *what* you pay, truthful bidding is weakly dominant. The "weakly" matters: against many rival profiles truth-telling only ties with other bids, so it is a safe prescription, not a unique prediction of what everyone else will do.

Verification. `check_C.py` checks the limit model as the second-price auction of item 23 (truthful bidding weakly dominates every other grid bid, with one or two rivals) and the one-cent-increment rule as described above. Tadelis gives no answer in the book.

Answer. (a) Bidding $b > v$ changes the outcome only when $v < h < b$, where it wins at a loss, so v weakly dominates b. (b) Bidding $b < v$ changes the outcome only when $b \leq h < v$, where it gives up a profitable win, so v weakly dominates b. (c) Proxy bidding is a second-price auction, so the true value is a weakly dominant bid, which justifies eBay's advice. With a positive increment, (b) holds only up to one increment: in cents, $v - 1$ cent weakly dominates v by half a cent.